

# Measure-to-measure interpolation using Transformers

A machine-checked blueprint: the original statements and proofs, our Lean formalization, and what it strengthens

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## About this blueprint

This is a standalone, A-to-Z blueprint of the controllability results of

M. Geshkovski, P. Rigollet, Y. Ruiz-Balet, *Measure-to-measure interpolation using Transformers*, arXiv:2411.04551v3,

together with their machine-checked Lean 4 formalization. It is meant to be read on its own: for each node it reproduces the **original statement (and proof, where short) verbatim**, then gives **our formalization** (a faithful statement in Lean, a prose proof, the real source listing, and an honest status badge), and, where relevant, **what our work strengthens** (a corrected sign error, a discharged rigor gap, or exactly what is axiomatised).

**Authorship and attribution.** The mathematics, the theorems, and the proofs are the work of the original authors. The Lean formalization, the numerical experiments, the prose informalizations, and this blueprint are the work of Alexandre Quemy. Original statements and proofs are reproduced verbatim, in clearly marked grey boxes with a page-cited attribution, for scholarly comparison (fair use); all such material remains the authors’.

### How to read the boxes.

**From the original paper.** Grey boxes reproduce the authors’ statements and proofs verbatim, with a page-cited attribution.

**Our formalization.** Blue boxes hold our Lean statement, a prose proof, the real source listing, and a status badge: **machine-checked** (clean Lean kernel proof), **axiomatised** (rests on a labeled axiom standing in for missing Mathlib theory), **open** (stated in Lean as a **sorry**), or **informal** (reviewed on paper).

**What we strengthen.** Green boxes record where the formalization corrects or completes the paper: a sign error (F1), a rigor gap discharged by a kernel-checked leaf (F2), or what had to be axiomatised.

The status badge of every machine-checked node is verified against the Lean kernel by `#print axioms (bin/axiom-report)`: a node is badged machine-checked only if its axiom set is clean (propext, Classical.choice, Quot.sound only).

# 1 Foundations

The paper views a Transformer as the flow map of a continuity equation on the unit sphere  $\mathbb{S}^{d-1}$ . We reproduce the setting verbatim, then give the kernel-checked algebraic foundations our leaves rest on.

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, §1, pp. 2-3).

Viewing each token as a particle, given an initial sequence  $(x_1(0), \dots, x_n(0)) \in (\mathbb{S}^{d-1})^n$ , one considers

$$\dot{x}_i(t) = \mathbf{v}[\mu(t)](t, x_i(t)), \quad t \in [0, T], \quad (1.1)$$

where  $\mu(t) = \frac{1}{n} \sum_{j=1}^n \delta_{x_j(t)}$  is the empirical measure. The vector field is

$$\mathbf{v}[\mu](t, x) = \mathbf{P}_x^\perp(\mathbf{V}(t) \mathcal{A}_{\mathbf{B}}[\mu](t, x) + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+), \quad (1.2)$$

where the self-attention is

$$\mathcal{A}_{\mathbf{B}}[\mu](t, x) := \int e^{\langle \mathbf{B}(t)x, x' \rangle} x' \mu(dx') / \int e^{\langle \mathbf{B}(t)x, \zeta \rangle} \mu(d\zeta).$$

The vector field is “ultimately projected onto  $\mathbb{T}_x \mathbb{S}^{d-1}$  by virtue of the orthogonal projector  $\mathbf{P}_x^\perp := I_d - xx^\top$ , referred to as *layer normalization*.” Since (1.1) depends only on  $\mu(t)$ , one passes to the **continuity equation**

$$\partial_t \mu(t) + \operatorname{div}(\mu(t) \mathbf{v}[\mu(t)]) = 0 \quad \text{on } [0, T] \times \mathbb{S}^{d-1}, \quad \mu(0) = \mu_0. \quad (1.3)$$

For any control  $\theta = (\mathbf{V}, \mathbf{B}, \mathbf{W}, \mathbf{U}, b)$  the Cauchy problem (1.3) is well-posed and yields a continuous, invertible flow map  $\Phi_\theta^t : \mathcal{P}(\mathbb{S}^{d-1}) \rightarrow \mathcal{P}(\mathbb{S}^{d-1})$  with  $\Phi_\theta^t(\mu_0) = \mu(t)$ .

## 1.1 Tangential projector (layer normalization)

**Our formalization.** machine-checked

We define the layer-normalization projector  $\mathbf{P}_x^\perp v = v - \langle x, v \rangle x$  and prove the algebraic identities every leaf uses (ledger node **L1**): it is linear, symmetric, idempotent on the sphere, annihilates  $x$ , and satisfies the quadratic identity

$$\langle \mathbf{P}_x^\perp v, v \rangle = \|v\|^2 - \langle x, v \rangle^2 \quad (\|x\| = 1),$$

which is exactly the scalar driving the gate ODE (B.5) and the barycenter ODE (B.9). The proof is a direct computation: expanding  $\langle v - \langle x, v \rangle x, v \rangle$  and using  $\langle x, x \rangle = 1$  cancels the cross term to  $\|v\|^2 - \langle x, v \rangle^2$ .

```

/-- The tangential projector  $P_{x^\perp} v = v - \langle x, v \rangle x$ . -/
noncomputable def tangentialProjector (x v : Eucl d) : Eucl d := v - (⟨x, v⟩) • x

/-- The key quadratic identity (L1): for a unit vector  $x$ ,
 $\langle P_{x^\perp} v, v \rangle = \|v\|^2 - \langle x, v \rangle^2$ . -/
theorem projector_inner_sub_sq {x : Eucl d} (hx : x ∈ sphere d) (v : Eucl d) :
  ⟨tangentialProjector x v, v⟩ = ‖v‖ ^ 2 - ⟨x, v⟩ ^ 2 := by
  have h : ⟨x, x⟩ = 1 := inner_self_eq_one_of_mem_sphere hx
  simp only [tangentialProjector, inner_sub_left, inner_smul_left, RCLike.conj_to_real,
    real_inner_self_eq_norm_sq]
  ring

```

[Foundations/Projector.lean](#)

## 1.2 Geodesic distance

**Our formalization.** machine-checked

The geodesic distance on the sphere is  $d_g(x, y) = \arccos \langle x, y \rangle$ . We prove the inversion  $\cos d_g(x, y) = \langle x, y \rangle$  for unit vectors and that  $d_g(x, y) \in [0, \pi]$ . These let the gate sign (B.4) be read either as the inner-product inequality  $\langle z, x \rangle < \cos R$  or as the geodesic statement  $d_g(z, x) > R$  (via strict antitonicity of  $\cos$  on  $[0, \pi]$ ).

```

/-- Geodesic distance on the unit sphere:  $d_g(x, y) = \arccos \langle x, y \rangle$ . -/
noncomputable def geodesicDist (x y : Eucl d) : ℝ := Real.arccos (⟨x, y⟩)

/--  $\cos (d_g x y) = \langle x, y \rangle$ : the cosine of the geodesic distance is the inner product.
    ↪ -/
theorem cos_geodesicDist {x y : Eucl d} (hx : x ∈ sphere d) (hy : y ∈ sphere d) :
  Real.cos (geodesicDist x y) = ⟨x, y⟩ :=
  Real.cos_arccos (neg_one_le_inner hx hy) (inner_le_one hx hy)

```

[Foundations/GeodesicDistance.lean](#)

## 2 Self-contained leaves

These are the self-contained computational cores the paper’s proofs rest on, each isolated from its measure-theoretic and flow-map scaffolding and given a clean Lean kernel proof. Two of them (L7, L8) reduce instead to a single labeled Wasserstein axiom and are flagged as such; the gate leaf (L2) and the non-colinearity leaves (L11, L11’) additionally repair or discharge a gap in the original text.

## 2.1 Gate algebra and gate ODE

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2 and eq. (B.4)-(B.5), pp. 31-32).

Lemma B.2 transports mass between two overlapping balls  $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$  ( $\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$ ): for any  $\varepsilon > 0$ ,  $T > 0$  there exist  $W, U \in \mathcal{M}_{d \times d}(\mathbb{R})$  and  $b \in \mathbb{R}^d$  so that the solution  $\mu$  satisfies  $\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1 - \varepsilon)\mu_0(\mathcal{B}_0)$ . In the proof:

“Let  $z \in \mathbb{S}^{d-1}$  denote the center and  $R > 0$  the radius of  $\mathcal{B}_0$ . Take an arbitrary  $\omega \in \text{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$ . We consider  $U = -\mathbf{1}z^\top$  and  $b = \cos(R)\mathbf{1}$ , as well as any  $W \in \mathcal{M}_{d \times d}(\mathbb{R})$  such that  $W\mathbf{1} = \omega$ . Then  $W(Ux + b)_+ = (-\cos d_g(z, x) + \cos(R))_+ \omega$ , and note that”

$$(-\cos d_g(z, x) + \cos(R))_+ > 0 \iff x \in \mathcal{B}_0. \quad (\text{B.4})$$

“Now observe that”

$$\frac{d}{dt} \langle x(t), \omega \rangle = (-\cos d_g(z, x(t)) + \cos(R))_+ (1 - \langle x(t), \omega \rangle^2), \quad (\text{B.5})$$

“which is positive whenever  $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$ .”

### Our formalization. machine-checked

The gated drift is  $W(Ux + b)_+ = g \cdot \omega$  with scalar gate  $g = (\cos R - \langle z, x \rangle)_+$ . Projecting onto the tangent space and pairing with the unit vector  $\omega$ , the projector identity L1 collapses the tangential part to a scalar:  $\langle \mathbf{P}_x^\perp(g\omega), \omega \rangle = g \langle \mathbf{P}_x^\perp \omega, \omega \rangle = g(\|\omega\|^2 - \langle x, \omega \rangle^2) = g(1 - \langle x, \omega \rangle^2)$  since  $\|\omega\| = 1$  — this is `gate_inner_identity`. The ODE (B.5) is then the chain rule:  $\frac{d}{dt} \langle x(t), \omega \rangle = \langle x'(t), \omega \rangle$  ( $\omega$  constant), and substituting the projected drift  $x'(t) = \mathbf{P}_{x(t)}^\perp(g\omega)$  reproduces the logistic factor. For the sign (B.4),  $0 < \cos R - \langle z, x \rangle \iff \langle z, x \rangle < \cos R$ , and rewriting  $\langle z, x \rangle = \cos d_g(z, x)$  with strict antitonicity of  $\cos$  on  $[0, \pi]$  gives the geodesic form. The key step is the quadratic projector identity, which turns the ambient projection into the scalar  $1 - \langle x, \omega \rangle^2$ .

```

/-- L2 (algebra of B.5): for unit `x` and unit `ω`, projecting the scaled drift `g•ω`
↪ and pairing
with `ω` gives the logistic factor `g(1 - ⟨x,ω⟩²)`. -/
theorem gate_inner_identity {x ω : Eucl d} (hx : x ∈ sphere d) (hω : ω ∈ sphere d) (g :
  ℝ) :
  ⟨tangentialProjector x (g • ω), ω⟩ = g * (1 - ⟨x, ω⟩ ^ 2) := by
  rw [tangentialProjector_smul, real_inner_smul_left, projector_inner_sub_sq hx ω,
    norm_eq_one_of_mem_sphere hω, one_pow]

/-- L2 (the gate ODE B.5): along a path `x(t)` on the sphere whose velocity is the
↪ projected gated
drift `x'(t) = P_{x(t)}^⊥ (g • ω)`, the inner product `⟨x(t), ω⟩` evolves as
`g(1 - ⟨x(t), ω⟩²)`. -/
theorem gate_hasDerivAt_inner {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
  (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hω : ω ∈ sphere d) (g : ℝ)
  (hode : x' = tangentialProjector (x t) (g • ω)) :
  HasDerivAt (fun s ⇒ (⟨x s, ω⟩ : ℝ)) (g * (1 - ⟨x t, ω⟩ ^ 2)) t := by
  have hconst : HasDerivAt (fun _ : ℝ ⇒ ω) (0 : Eucl d) t := hasDerivAt_const t ω
  have h := hx.inner ℝ hconst
  rw [hode, gate_inner_identity hxs hω g] at h
  simpa using h

/-- L2 (gate sign, B.4, geodesic form): the gate is active exactly outside the closed
↪ cap of radius
`R` around `z`, i.e. on `{x : d_g(z, x) > R}`. -/
theorem gate_pos_iff_dist {z x : Eucl d} (hz : z ∈ sphere d) (hx : x ∈ sphere d)
  {R : ℝ} (hR : R ∈ Set.Icc (0 : ℝ) Real.pi) :
  (⟨z, x⟩ : ℝ) < Real.cos R ↔ R < geodesicDist z x := by
  rw [← cos_geodesicDist hz hx]
  exact Real.strictAntiOn_cos.lt_iff_gt (geodesicDist_mem_Icc z x) hR

```

[Leaves/GateODE.lean](#)

**Strengthening (finding F1).** The printed (B.4) asserts the gate is active iff  $x \in \mathcal{B}_0 = B(z, R)$ . Our kernel-checked `gate_pos_iff_dist` proves instead  $(\cos R - \langle z, x \rangle)_+ > 0 \iff d_g(z, x) > R$  — the gate is active on the **complement** of the cap. The opposite sign is the consistent one: (B.5) needs the gate active *inside*  $\mathcal{B}_0 \setminus \{\omega\}$  to push interior mass toward  $\omega$ , whereas the printed parameters  $U = -\mathbf{1}z^\top$ ,  $b = \cos(R)\mathbf{1}$  leave the interior mass on a zero gate, so the lemma as written transports nothing; the fix is  $U = +\mathbf{1}z^\top$ ,  $b = -\cos(R)\mathbf{1}$ . This is corroborated independently by experiment E1 (the gated-transport mass fraction rises from  $\approx 0.27$  to 1.0 once the seed is placed on  $\{d_g(z, x) > R\}$ ) and by the paper’s own §4 Step 3, which states the identical construction with the correct side. See `ERRATA.md`.

## 2.2 Barycenter ODE

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, eq. (B.9), p. 33).

In Step 1 of the proof of Lemma 3.3, the solution to

$$\begin{cases} \partial_t \mu(t) + \operatorname{div}(\mathbf{P}_x^\perp \langle \alpha_1, \mathbb{E}_{\mu(t)}[x] \rangle \alpha_1 \mu(t)) = 0 & \text{on } \mathbb{R}_{\geq 0} \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1} \end{cases} \quad (\text{B.9})$$

for  $\mu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$  satisfies

$$\frac{d}{dt} \langle \mathbb{E}_{\mu(t)}[x], \alpha_1 \rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1 \rangle \left( 1 - \int \langle x', \alpha_1 \rangle^2 \mu(t, dx') \right).$$

**Our formalization.** machine-checked

The disentangling field is  $\mathbf{P}_x^\perp(c\alpha)$  with  $c = \langle \mathbb{E}_\mu[x], \alpha \rangle$ , so the per-particle characteristic  $\langle x(t), \alpha \rangle$  obeys exactly the gate identity of L2 with  $c$  in place of the gate scalar:  $\frac{d}{dt} \langle x(t), \alpha \rangle = c(1 - \langle x(t), \alpha \rangle^2)$ . The Lean derivative is therefore literally `gate_hasDerivAt_inner` instantiated at  $g = c$ . The sign lemma `barycenter_deriv_pos` records that when  $c > 0$  and the component lies strictly inside  $(-1, 1)$  the right-hand side is positive, the dissipation that drives  $\langle x, \alpha \rangle \rightarrow 1$ , i.e.  $x \rightarrow \alpha$ . The key point is that the paper shows  $c$  is sign-preserved along the flow, so the logistic ODE is monotone.

```
/-- L6 (derivative, eq. B.9): along `x'(t) = P_{x(t)}^\perp (c \cdot \alpha)` with `alpha` a unit vector,
  ↪ the
component ` $\langle x(t), \alpha \rangle$ ` evolves as ` $c(1 - \langle x(t), \alpha \rangle^2)$ `. -/
theorem barycenter_hasDerivAt_inner {x : ℝ → Eucl d} {α : Eucl d} {t : ℝ} {x' : Eucl d}
  (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hα : α ∈ sphere d) (c : ℝ)
  (hode : x' = tangentialProjector (x t) (c • α)) :
  HasDerivAt (fun s => (⟨x s, α⟩ : ℝ)) (c * (1 - ⟨x t, α⟩ ^ 2)) t :=
  gate_hasDerivAt_inner hx hxs hα c hode

/-- L6 (sign / strict increase): when the barycenter coefficient `c` is positive and the
  ↪ component
is strictly inside `(-1, 1)`, the component strictly increases. This is the dissipation
  ↪ that pushes
`⟨x, α⟩` toward `1`. -/
theorem barycenter_deriv_pos {c y : ℝ} (hc : 0 < c) (hy : y ∈ Set.Ioo (-1 : ℝ) 1) :
  0 < c * (1 - y ^ 2) := by
  obtain ⟨h1, h2⟩ := hy
  have hy2 : y ^ 2 < 1 := by nlinarith
  have : 0 < 1 - y ^ 2 := by linarith
  positivity
```

[Leaves/BarycenterODE.lean](#)

## 2.3 Geodesic-distance gradient

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, eq. after (4.4), p. 20).

“Observe that the trajectories of the ODE”

$$\dot{x}(t) = (\langle \gamma, x(t) \rangle - \epsilon/2)_+ \mathbf{P}_{x(t)}^\perp \omega_+ \quad \text{for } t \geq 0, \quad (4.4)$$

“follow the Riemannian gradient flow of the distance between  $\omega_+$  and  $x$  in  $\mathcal{S}_+$ . Indeed,”

$$\nabla_1 d_g(x, \omega_+) = -\frac{\mathbf{P}_x^\perp(\omega_+)}{\sqrt{1 - \langle x, \omega_+ \rangle^2}}.$$

**Our formalization.** machine-checked

Two facts pin down the gradient. First, the chain rule for  $\arccos \circ \langle \cdot, \omega \rangle$ : since  $(\arccos)'(u) = -1/\sqrt{1-u^2}$  (Mathlib’s `Real.hasDerivAt_arccos`),  $\frac{d}{dt}d_g(x(t), \omega) = -(1 - \langle x, \omega \rangle^2)^{-1/2} \langle x', \omega \rangle$ . Second, for a tangent velocity ( $\langle x', x \rangle = 0$ ) the ambient  $\omega$  may be replaced by its tangential projection:  $\langle x', \mathbf{P}_x^\perp \omega \rangle = \langle x', \omega \rangle - \langle x, \omega \rangle \langle x', x \rangle = \langle x', \omega \rangle$ . Hence the derivative is  $\langle x', -(1 - \langle x, \omega \rangle^2)^{-1/2} \mathbf{P}_x^\perp \omega \rangle$ , exhibiting  $-\mathbf{P}_x^\perp \omega / \sqrt{1 - \langle x, \omega \rangle^2}$  as the Riemannian (tangential) gradient. The key step is that tangency annihilates the cross term, letting the Euclidean inner product against  $\omega$  be read as the inner product against the gradient direction.

```
/-- L4 (derivative of geodesic distance along a path): the chain rule for `arccos ∘ ⟨·, ω⟩`,
↪ ω⟩`. -/
theorem geodesicDist_hasDerivAt {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
  (hx : HasDerivAt x x' t) (h1 : (⟨x t, ω⟩ : ℝ) ≠ -1) (h2 : (⟨x t, ω⟩ : ℝ) ≠ 1) :
  HasDerivAt (fun s => geodesicDist (x s) ω)
    (-(1 / Real.sqrt (1 - (⟨x t, ω⟩ : ℝ) ^ 2)) * ⟨x', ω⟩) t := by
  have hinner : HasDerivAt (fun s => (⟨x s, ω⟩ : ℝ)) (⟨x', ω⟩) t := by
    have h := hx.inner ℝ (hasDerivAt_const t ω)
    simpa using h
  have h := (Real.hasDerivAt_arccos h1 h2).comp t hinner
  simpa only [geodesicDist, Function.comp_def] using h

/-- L4 (gradient direction): on the sphere, for a tangent velocity the inner product
↪ with `ω` equals
the inner product with the tangential projection `P_x^⊥ ω`. Hence the geodesic-distance
↪ derivative
is `⟨x', -(1/√(1-⟨x,ω⟩²)) • P_x^⊥ ω⟩`, exhibiting `-P_x^⊥ ω / √(1-⟨x,ω⟩²)` as the
↪ Riemannian
gradient. -/
theorem inner_eq_inner_tangentialProjector {x ω x' : Eucl d}
  (htangent : (⟨x', x⟩ : ℝ) = 0) :
  (⟨x', ω⟩ : ℝ) = ⟨x', tangentialProjector x ω⟩ := by
  rw [tangentialProjector, inner_sub_right, real_inner_smul_right, htangent]
  ring
```

[Leaves/GeodesicGradient.lean](#)



## 2.4 Separating hyperplane

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 4.2 Step 1, p. 20).

With  $\omega_+$  on the minimizing geodesic between  $\omega$  and  $\gamma$  at distance  $\pi/8$  from  $\omega$ , one has  $d_g(\omega_+, x_0^M) \geq d_g(\omega, x_0^M) - d_g(\omega, \omega_+) \geq \frac{3\pi}{8}$ .

“As a consequence, the hyperplane  $\{x \in \mathbb{S}^{d-1} : \langle \omega, x \rangle = \cos(\pi/8 + \tau)\}$  is a separating hyperplane for the ball  $B(\omega, \pi/8 + \tau)$  and the points  $x_0^M$  and  $y^M$  for every  $\tau \in (0, 3\pi/8)$ ; namely”

$$\langle \omega, x_0^M \rangle - \cos\left(\frac{\pi}{8} + \tau\right) = \cos d_g(\omega, x_0^M) - \cos\left(\frac{\pi}{8} + \tau\right) < 0,$$

“where the inequality is by virtue of (4.3)” (the choice  $d_g(\omega, x_0^M) \geq \pi/2$ ).

**Our formalization.** machine-checked

The whole content is the strict antitonicity of cosine on  $[0, \pi]$ . From  $\tau < 3\pi/8$  we get the cap angle  $\pi/8 + \tau < \pi/2 \leq d_g(\omega, x)$ , and both angles lie in  $[0, \pi]$ , so Mathlib’s `Real.strictAntiOn_cos` gives  $\cos d_g(\omega, x) < \cos(\pi/8 + \tau)$ . Rewriting  $\cos d_g(\omega, x) = \langle \omega, x \rangle$  lands the separating inequality  $\langle \omega, x \rangle < \cos(\pi/8 + \tau)$ . The key step is that monotonicity of cosine converts the angle gap  $\pi/8 + \tau < d_g(\omega, x)$  into the inner-product gap that places  $x$  strictly on the far side of the hyperplane.

```
-- L3 (separating side): if 'x' is far from the anchor 'ω' ('d_g(ω, x) ≥ π/2') and
-- 'τ ∈ (0, 3π/8)', then '⟨ω, x⟩ < cos(π/8 + τ)', so 'x' lies strictly on the far side of
-- the
separating hyperplane. -/
theorem separating_hyperplane {x ω : Eucl d} (hx : x ∈ sphere d) (hw : ω ∈ sphere d)
  {τ : ℝ} (ht : τ ∈ Set.Ioo 0 (3 * Real.pi / 8))
  (hfar : Real.pi / 2 ≤ geodesicDist ω x) :
  ⟨ω, x⟩ < Real.cos (Real.pi / 8 + τ) := by
  obtain ⟨ht0, ht8⟩ := ht
  have hpi := Real.pi_pos
  -- both angles lie in [0, π]
  have ha : Real.pi / 8 + τ ∈ Set.Icc (0 : ℝ) Real.pi := by
    constructor <=> nlinarith [Real.pi_pos]
  have hb : geodesicDist ω x ∈ Set.Icc (0 : ℝ) Real.pi := geodesicDist_mem_Icc ω x
  -- the far angle exceeds the cap angle
  have hab : Real.pi / 8 + τ < geodesicDist ω x := by nlinarith [hfar]
  -- cosine is strictly decreasing on [0, π]
  have hcos : Real.cos (geodesicDist ω x) < Real.cos (Real.pi / 8 + τ) :=
    Real.strictAntiOn_cos ha hb hab
  rwa [cos_geodesicDist hw hx] at hcos
```

[Leaves/SeparatingHyperplane.lean](#)

## 2.5 Lyapunov sign

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Example 6.1, pp. 29-30).

On the circle ( $d = 2$ ), for a particle following a characteristic of  $\mu^i(t)$  one has  $\dot{x}(t) = \mathbf{P}_{x(t)}^\perp(m_i(t)) = \alpha_i(t)\mathbf{P}_{x(t)}^\perp(u_i)$  and, in the angle variable  $\vartheta := \theta - \theta_i$ ,

$$\dot{\vartheta}(t) = -\alpha_i(t) \sin \vartheta(t).$$

“The Lyapunov function  $E(\vartheta) := 1 - \cos \vartheta$  satisfies”

$$\dot{E} = \sin \vartheta \dot{\vartheta} = -\alpha_i(t) \sin^2 \vartheta \leq 0,$$

“with equality only at  $\vartheta = 0$  (since  $\alpha_i(t) > 0$  ...). Thus  $\vartheta(t) \rightarrow 0$  and each trajectory converges to  $u_i$ .”

**Our formalization.** machine-checked

Both halves are kernel-checked. The derivative is the chain rule:  $\frac{d}{dt}(1 - \cos \theta(s)) = \sin \theta(t) \theta'(t)$ , and substituting the ODE  $\theta' = -\alpha \sin \theta$  gives  $-\alpha \sin^2 \theta$ . The sign lemma `lyapunov_deriv_nonpos` then notes  $-\alpha \sin^2 \theta \leq 0$  for  $\alpha \geq 0$  because  $\alpha \sin^2 \theta \geq 0$ . So  $E = 1 - \cos \theta$  is nonincreasing along the angle flow, forcing  $\theta \rightarrow 0$ . The key step is that the  $\sin \theta$  produced by the ODE cancels with the  $\sin \theta$  from differentiating  $1 - \cos \theta$ , leaving a perfect square of definite sign.

```
-- L5 (derivative of the Lyapunov function along the flow): if `θ` solves `θ' = -α sin
  ↳ θ` at `t`,
then `E(s) = 1 - cos θ(s)` has derivative `-α sin²(θ t)` at `t`. -/
theorem lyapunov_hasDerivAt (α : ℝ) (θ : ℝ → ℝ) (t θ' : ℝ)
  (hθ : HasDerivAt θ θ' t) (hode : θ' = -α * Real.sin (θ t)) :
  HasDerivAt (fun s => 1 - Real.cos (θ s)) (-α * Real.sin (θ t) ^ 2) t := by
  have hcos : HasDerivAt (fun s => Real.cos (θ s)) (-Real.sin (θ t) * θ') t :=
    (Real.hasDerivAt_cos (θ t)).comp t hθ
  have h : HasDerivAt (fun s => 1 - Real.cos (θ s)) (0 - (-Real.sin (θ t) * θ')) t :=
    (hasDerivAt_const t 1).sub hcos
  convert h using 1
  rw [hode]; ring

-- L5 (sign): the Lyapunov derivative is nonpositive when `α ≥ 0`. -/
theorem lyapunov_deriv_nonpos {α : ℝ} (hα : 0 ≤ α) (s : ℝ) :
  -α * Real.sin s ^ 2 ≤ 0 := by
  have : 0 ≤ α * Real.sin s ^ 2 := mul_nonneg hα (sq_nonneg _)
  linarith
```

[Leaves/Lyapunov.lean](#)

## 2.6 Ball-chain retention

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1 and its proof, pp. 31, 33).

Lemma B.1 transports mass through  $K + 1$  overlapping balls and concludes  $\mu(T, \mathcal{B}_K) \geq (1 - \varepsilon)^K \mu_0(\bigcup_k \mathcal{B}_k)$ . Its proof:

“Write  $[0, T] = \bigcup_{k \in [1, M]} [t_{k-1}, t_k]$  where  $t_k = kT/K$ , and proceed by backward induction:”

$$\mu(T, \mathcal{B}_K) = \mu(T, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T, \mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1}, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1 - \varepsilon) \mu(t_{K-1}, \mathcal{B}_{K-1}),$$

“where the last inequality follows from Lemma B.2. Using  $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$  whenever  $|k - k'| \geq 2$ , we arrive to”

$$\mu(T, \mathcal{B}_K) \geq (1 - \varepsilon)^K \left( \sum_{k=1}^K \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0) \right) = (1 - \varepsilon)^K \mu_0 \left( \bigcup_{k \in [0, K]} \mathcal{B}_k \right).$$

**Our formalization.** machine-checked

Stripped of the measure-theoretic bookkeeping, the backward induction is a clean arithmetic fact about a nonnegative sequence with per-step retention. We prove by induction on  $K$  that if  $m \leq a_0$  and  $(1 - \varepsilon) a_k \leq a_{k+1}$  for all  $k$ , then  $(1 - \varepsilon)^K m \leq a_K$ . The base case is  $m \leq a_0$ ; the step chains  $(1 - \varepsilon)^{n+1} m = (1 - \varepsilon)((1 - \varepsilon)^n m) \leq (1 - \varepsilon) a_n \leq a_{n+1}$ . The key step is  $0 \leq 1 - \varepsilon$ , which lets the induction hypothesis be scaled by  $(1 - \varepsilon)$  without reversing the inequality before applying the retention hypothesis.

```
/-- L9: geometric retention through a chain. With retention factor `1 - ε ∈ [0, 1]` per
    ↳ step and
initial mass at least `m ≥ 0`, after `K` steps at least `(1 - ε)^K · m` remains. -/
theorem ball_chain_geom {ε : ℝ} (hε1 : ε ≤ 1) (a : ℕ → ℝ) {m : ℝ}
  (h0 : m ≤ a 0) (hstep : ∀ k, (1 - ε) * a k ≤ a (k + 1)) :
  ∀ K : ℕ, (1 - ε) ^ K * m ≤ a K := by
  have h1ε : (0 : ℝ) ≤ 1 - ε := by linarith
  intro K
  induction K with
  | zero => simp using h0
  | succ n ih =>
    calc (1 - ε) ^ (n + 1) * m
      = (1 - ε) * ((1 - ε) ^ n * m) := by ring
      ≤ (1 - ε) * a n := by exact mul_le_mul_of_nonneg_left ih h1ε
      ≤ a (n + 1) := hstep n
```

[Leaves/BallChain.lean](#)

## 2.7 Pigeonhole

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4 Part 1, App. B.3, p. 35).

“Part 1. There exists an open ball  $\mathcal{B} \subset \text{supp } \mu_0 \cup \text{supp } \nu_0$  such that  $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$ . We now claim that there exists some  $x^* \in \mathcal{B}$  such that”

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x \mu_0(dx) \neq \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x \nu_0(dx).$$

“Indeed if this were to be false, then we’d have”

$$x^* = \frac{1}{\mu_0(\mathcal{B}) - \nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x (\nu_0(dx) - \mu_0(dx))$$

“for all  $x^* \in \mathcal{B}$ , which cannot hold.”

**Our formalization.** machine-checked

The displayed contradiction is “ $x^*$  equals a fixed vector for all  $x^* \in \mathcal{B}$ ” — i.e. the identity would be constant on the open ball  $\mathcal{B}$ , impossible. We isolate exactly that general fact: in a nontrivial real normed space, an open ball of positive radius contains a point distinct from any prescribed  $a$ . Pick  $u \neq 0$  and set  $v = (r/2\|u\|) u$ , so  $\|v\| = r/2 < r$  and  $v \neq 0$ ; then  $c$  and  $c + v$  both lie in  $B(c, r)$  (at distances 0 and  $r/2$ ) and are distinct. If  $c = a$ , return  $c + v$  (which differs from  $c = a$  because  $v \neq 0$ ); otherwise return  $c$ . The key step is that a positive-radius ball is never a single point, so it cannot be forced to one value — the contradiction the pigeonhole argument exploits.

```

/-- L10: an open ball of positive radius in a nontrivial real normed space contains a
↪ point distinct
from any prescribed 'a'. Hence no map can be forced to a single value on the ball, which
↪ is the
contradiction the pigeonhole step exploits. -/
theorem exists_ne_in_ball (c a : E) {r : ℝ} (hr : 0 < r) :
  ∃ x ∈ Metric.ball c r, x ≠ a := by
  obtain ⟨u, hu⟩ := exists_ne (0 : E)
  have hunorm : 0 < ‖u‖ := norm_pos_iff.mpr hu
  set v : E := (r / (2 * ‖u‖)) • u with hv
  have hvnorm : ‖v‖ = r / 2 := by
    rw [hv, norm_smul, Real.norm_of_nonneg (by positivity)]
    field_simp
  have hvne : v ≠ 0 := by
    rw [← norm_pos_iff, hvnorm]; linarith
  -- 'c' and 'c + v' both lie in the ball and are distinct
  have hc_mem : c ∈ Metric.ball c r := Metric.mem_ball_self hr
  have hcv_mem : c + v ∈ Metric.ball c r := by
    rw [Metric.mem_ball, dist_eq_norm]
    have h1 : c + v - c = v := by abel
    rw [h1, hvnorm]; linarith
  -- at least one of the two distinct points differs from 'a'
  rcases eq_or_ne c a with hca | hca
  · refine ⟨c + v, hcv_mem, fun h ⇒ hvne ?_⟩
    have h2 : c + v = c + 0 := by rw [add_zero, h, hca]
    exact add_left_cancel h2
  · exact ⟨c, hc_mem, hca⟩

```

[Leaves/Pigeonhole.lean](#)

## 2.8 Disjoint hulls give non-colinear barycenters

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).

The induction hypothesis (3.3) is

$$\text{conv}_g \text{supp } \mu_0^i \cap \text{conv}_g \text{supp } \mu_0^j = \emptyset \quad \text{for } i \neq j \in \llbracket 1, N-1 \rrbracket. \quad (3.3)$$

The induction step then asserts, without proof:

“Since  $\text{supp } \mu_0^i \subset \mathbb{Q}_1^{d-1}$ , (3.3) implies that”

$$\mathbb{E}_{\mu_0^i}[x] \text{ is not colinear with } \mathbb{E}_{\mu_0^j}[x] \quad \text{for } i \neq j \in \llbracket 1, N-1 \rrbracket.$$

**Our formalization.** machine-checked

Within an open hemisphere spherical geodesics are radial projections of Euclidean chords, so we model the geodesic hull as  $\text{conv}_g(s) = \text{cone}(s) \cap \mathbb{S}^{d-1}$  — the unit vectors that are nonnegative combinations of  $s$  (`geodesicHull`). The empirical barycenter  $\sum_p w_p p$  with  $w_p \geq 0$  is a

manifest conical combination of  $s$ , and the cone is closed under nonnegative scaling, so the normalization  $\|b\|^{-1}b$  of any nonzero conical point lies in  $\text{conv}_g(s)$ . If the two barycenters were colinear — for vectors in the positive orthant “colinear” means a *positive* multiple, i.e. **SameRay** — then  $b = k c$  with  $k > 0$ , so their normalizations coincide,  $\|b\|^{-1}b = \|c\|^{-1}c$ . That common unit direction would then sit in *both* geodesic hulls, contradicting their disjointness. The key step is that positive colinearity forces a shared normalized direction, which disjoint hulls forbid.

```

/-- The geodesic-convex hull of `s`, modeled within a hemisphere as `cone(s) ∩ S^{d-1}`:
  ↳ the unit
vectors that are nonnegative combinations of `s`. Faithful to `conv_g` when `s` lies in
  ↳ an open
hemisphere (e.g. the orthant `Q_1^{d-1}`), since spherical geodesics are radial
  ↳ projections of chords. -/
def geodesicHull (s : Finset (Eucl d)) : Set (Eucl d) :=
  {x | ||x|| = 1 ∧ inConicalSpan s x}

/-- **Core of finding F2.** If two nonzero conical points `b`, `c` (the unnormalized
  ↳ barycenters) are
colinear - `SameRay ℝ b c`, i.e. positive multiples, which is what "colinear" means for
  ↳ vectors in the
positive orthant - then their common normalized direction lies in both geodesic hulls.
  ↳ -/
theorem geodesicHull_inter_nonempty_of_sameRay {s₁ s₂ : Finset (Eucl d)} {b c : Eucl d}
  (hb : inConicalSpan s₁ b) (hc : inConicalSpan s₂ c) (hb0 : b ≠ 0) (hc0 : c ≠ 0)
  (hsr : SameRay ℝ b c) : (geodesicHull s₁ ∩ geodesicHull s₂).Nonempty := by
  obtain ⟨r₁, r₂, hr₁, hr₂, hrel⟩ := hsr.exists_pos hb0 hc0
  -- `b = k • c` with `k = r₁⁻¹ r₂ > 0`
  set k : ℝ := r₁⁻¹ * r₂ with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr₁), one_smul]
  -- the normalized vectors coincide
  have key : ||b||⁻¹ • b = ||c||⁻¹ • c := by
    have hnc : ||c|| ≠ 0 := norm_ne_zero_iff.mpr hc0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  refine ⟨||b||⁻¹ • b, ?_⟩
  rw [Set.mem_inter_iff]
  exact ⟨mem_geodesicHull_normalize hb hb0, key ▸ mem_geodesicHull_normalize hc hc0⟩

/-- **Finding F2, discharged.** Empirical measures (with nonnegative weights) in the
  ↳ orthant whose
geodesic hulls are disjoint have non-colinear barycenters. This is the step Proposition
  ↳ 3.1's
induction asserts without proof. -/
theorem barycenter_noncolinear_of_disjoint_hull {s₁ s₂ : Finset (Eucl d)} {w₁ w₂ : Eucl
  ↳ d → ℝ}
  (hw₁ : ∀ p ∈ s₁, 0 ≤ w₁ p) (hw₂ : ∀ p ∈ s₂, 0 ≤ w₂ p)
  (hb0 : (∑ p ∈ s₁, w₁ p • p) ≠ 0) (hc0 : (∑ p ∈ s₂, w₂ p • p) ≠ 0)
  (hdisj : Disjoint (geodesicHull s₁) (geodesicHull s₂)) :
  ¬ SameRay ℝ (∑ p ∈ s₁, w₁ p • p) (∑ p ∈ s₂, w₂ p • p) := by
  intro hsr
  obtain ⟨u, hu⟩ := geodesicHull_inter_nonempty_of_sameRay
    (barycenter_mem_conicalSpan hw₁) (barycenter_mem_conicalSpan hw₂) hb0 hc0 hsr
  rw [Set.mem_inter_iff] at hu
  exact Set.disjoint_left.mp hdisj hu.1 hu.2

```

Leaves/BarycenterNonColinear.lean

**Strengthening (finding F2).** Proposition 3.1’s induction *asserts* “(3.3) [disjoint geodesic hulls]  $\Rightarrow \mathbb{E}_{\mu_0^i}[x]$  not colinear with  $\mathbb{E}_{\mu_0^j}[x]$ ” with no justification — the one genuine rigor gap the adversarial review flagged. This leaf supplies the kernel-checked proof for the empirical regime (the Dirac targets of Theorem 1.1 and the empirical targets of the restricted Theorem 1.2), under the faithful hemisphere model  $\text{conv}_g(s) = \text{cone}(s) \cap \mathbb{S}^{d-1}$ .

## 2.9 F2, general probability measures

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).

The same unproved assertion as above, now read for arbitrary  $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$  with barycenter  $\mathbb{E}_{\mu_0}[x] = \int x \mu_0(dx)$ :

$$\text{conv}_g \text{supp } \mu_0 \cap \text{conv}_g \text{supp } \nu_0 = \emptyset \implies \mathbb{E}_{\mu_0}[x] \text{ not colinear with } \mathbb{E}_{\nu_0}[x].$$

**Our formalization.** machine-checked

The finite case above is elementary; for a general probability measure the only extra ingredient is the standard fact that the barycenter  $\int x d\mu$  lies in any closed convex set carrying full mass — Mathlib’s `Convex.integral_mem`. Wrapping it as `barycenter_mem_of_supportedIn` (from  $\mu(s^c) = 0$  to  $\int x d\mu \in s$ ), the same `SameRay` argument runs verbatim: both normalized barycenters land in their respective closed convex *cones*  $s_1, s_2$ , the shared unit direction sits in both, and disjointness on the sphere is contradicted. The key step is that `Convex.integral_mem` is the measure-theoretic analogue of “the barycenter lies in the convex hull of the support,” so no optimal-transport axiom is reintroduced.



```

/-- The barycenter of a probability measure lies in any closed convex set carrying full
    ↪ mass
( $\mu S^c = 0$ ). This is the standard measure-theoretic fact the general case of finding F2
    ↪ needs; it
rests only on Mathlib's 'Convex.integral_mem', not on the optimal-transport axioms. -/
theorem barycenter_mem_of_supportedIn { $\mu$  : Measure (Eucl d)} [IsProbabilityMeasure  $\mu$ ]
  (hint : Integrable (fun x : Eucl d  $\Rightarrow$  x)  $\mu$ ) {s : Set (Eucl d)}
  (hconv : Convex  $\mathbb{R}$  s) (hclosed : IsClosed s) (hsupp :  $\mu S^c = 0$ ) :
  barycenter  $\mu \in s$  := by
  have hae :  $\forall^m x \partial\mu, x \in s$  := by
    rw [ae_iff]; exact hsupp
  rw [barycenter]
  exact hconv.integral_mem hclosed hae hint

/-- **Finding F2, general case.** Two probability measures whose supports lie in
    ↪ disjoint closed
convex cones 's1', 's2' (disjoint on the sphere) have non-colinear barycenters.
    ↪ Generalizes
'barycenter_noncolinear_of_disjoint_hull' from empirical to arbitrary probability
    ↪ measures; kernel-
clean, via 'barycenter_mem_of_supportedIn'. -/
theorem barycenter_noncolinear_of_disjoint_hull_general
  { $\mu \nu$  : Measure (Eucl d)} [IsProbabilityMeasure  $\mu$ ] [IsProbabilityMeasure  $\nu$ ]
  (h $\mu$ i : Integrable (fun x : Eucl d  $\Rightarrow$  x)  $\mu$ ) (h $\nu$ i : Integrable (fun x : Eucl d  $\Rightarrow$  x)  $\nu$ )
  {s1 s2 : Set (Eucl d)}
  (hc1 : Convex  $\mathbb{R}$  s1) (hcl1 : IsClosed s1) (hcone1 :  $\forall x \in s_1, \forall a : \mathbb{R}, 0 \leq a \rightarrow a \cdot x \in s_1$ )
    ↪ s1)
  (hc2 : Convex  $\mathbb{R}$  s2) (hcl2 : IsClosed s2) (hcone2 :  $\forall x \in s_2, \forall a : \mathbb{R}, 0 \leq a \rightarrow a \cdot x \in s_2$ )
    ↪ s2)
  (h $\mu$ s :  $\mu S_1^c = 0$ ) (h $\nu$ s :  $\nu S_2^c = 0$ )
  (hb0 : barycenter  $\mu \neq 0$ ) (h $\nu$ 0 : barycenter  $\nu \neq 0$ )
  (hdisj :  $\forall u : \text{Eucl } d, \|u\| = 1 \rightarrow u \in s_1 \rightarrow u \notin s_2$ ) :
   $\neg \text{SameRay } \mathbb{R} (\text{barycenter } \mu) (\text{barycenter } \nu)$  := by
  intro hsr
  set b := barycenter  $\mu$  with hbdef
  set c := barycenter  $\nu$  with hcdef
  obtain (r1, r2, hr1, hr2, hrel) := hsr.exists_pos hb0 h $\nu$ 0
  have hbs1 : b  $\in s_1$  := barycenter_mem_of_supportedIn h $\mu$ i hc1 hcl1 h $\mu$ s
  have hcs2 : c  $\in s_2$  := barycenter_mem_of_supportedIn h $\nu$ i hc2 hcl2 h $\nu$ s
  set k :  $\mathbb{R}$  := r1-1 * r2 with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr1), one_smul]
  have key :  $\|b\|^{-1} \cdot b = \|c\|^{-1} \cdot c$  := by
    have hnc :  $\|c\| \neq 0$  := norm_ne_zero_iff.mpr h $\nu$ 0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  set u :=  $\|b\|^{-1} \cdot b$  with hu
  have hunorm :  $\|u\| = 1$  := by
    rw [hu, norm_smul, norm_inv, norm_norm, inv_mul_cancel₀ (norm_ne_zero_iff.mpr hb0)]
  have hus1 : u  $\in s_1$  := by rw [hu]; exact hcone1 b hbs1 _ (by positivity)
  have hus2 : u  $\in s_2$  := by rw [key]; exact hcone2 c hcs2 _ (by positivity)
  exact hdisj u hunorm hus1 hus2

```

Leaves/BarycenterNonColinear.lean

**Strengthening (finding F2, residual).** The finite leaf above closes F2 for the empirical regime; this companion closes the remaining general-measure case, extending the non-collinearity to arbitrary probability measures supported in disjoint closed convex cones. It rests only on Mathlib’s `Convex.integral_mem` and introduces **no new axioms** — in particular it does not reach for the optimal-transport ( $W_2$ ) axiom.

## 2.10 Linearized OT bound

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.2, p. 24).

“Lemma 5.2. Suppose  $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$  and  $T^1, T^2 : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  measurable, with  $T^1$  bijective. Then”

$$W_2(T^1_{\#}\mu, T^2_{\#}\mu) \lesssim \|T^1 - T^2\|_{L^2(\mu)}. \quad (5.2)$$

“Proof of Lemma 5.2. Since  $T^1$  is bijective, there is a measurable  $\psi : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  such that  $\psi(T^1(x)) = T^2(x)$  for all  $x \in \mathbb{S}^{d-1}$ . Then”

$$W_2^2(T^1_{\#}\mu, T^2_{\#}\mu) \lesssim \int \|x - \psi(x)\|^2 (T^1_{\#}\mu)(dx) = \|T^1 - T^2\|_{L^2(\mu)}^2.$$

**Our formalization.** axiomatised

This is the map-induced coupling bound. Since  $T^1$  is bijective there is  $\psi$  with  $\psi \circ T^1 = T^2$ , and the deterministic coupling  $(\text{id}, \psi)_{\#}(T^1_{\#}\mu)$  between the two pushforwards realizes transport cost  $\int \|x - \psi(x)\|^2 d(T^1_{\#}\mu) = \|T^1 - T^2\|_{L^2(\mu)}^2$ , bounding  $W_2^2$ . Mathlib **v4.31.0** has no  $W_2$ , so the Lean statement is asserted over the labeled optimal-transport axiom `W2_map_le_L2`; the node is therefore **axiomatised**, not a kernel proof. Its effective status is the minimum over its dependency closure, which contains the  $W_2$  axiom.

```
-- L7 (Lemma 5.2): the `W2` distance between two pushforwards of `μ` is bounded by the
↪ `L^2(μ)`
distance of the maps. Rests on the optimal-transport axiom `W2_map_le_L2`. -/
theorem lemma_5_2 (μ : Measure (Eucl d)) (T1 T2 : Eucl d → Eucl d) :
  W2 (μ.map T1) (μ.map T2) ≤ Real.sqrt (∫ x, ‖T1 x - T2 x‖ ^ 2 ∂μ) :=
  W2_map_le_L2 μ T1 T2
```

Leaves/Coupling.lean

## 2.11 Markov bound

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Claim 2, App. B.5, p. 39).

“Claim 2. Suppose  $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$  and  $x_0 \in \mathbb{S}^{d-1}$  satisfy  $W_2(\mu, \delta_{x_0}) \leq \eta_2$ . There exists a universal constant  $C > 0$  such that  $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$  for all  $\eta_3 > 0$ .”

“Proof of Claim 2. By compactness of  $\mathbb{S}^{d-1}$  and Kantorovich-Rubinstein duality,”

$$W_1(\mu, \delta_{x_0}) = \sup_{\text{Lip}(g) \leq 1} \int g(\mu - \delta_{x_0}) \leq C \cdot \eta_2$$

“for some numerical constant  $C > 0$ . Hence, for  $g : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  defined as”

$$g(x) = \begin{cases} 1 & x \in B(x, \eta_3) \\ 1 - \frac{1 - \eta_3}{\eta_3} & x \in B(x, \eta_3) \cap B(x, 2\eta_3) \\ 0 & x \notin B(x, 2\eta_3), \end{cases}$$

“we obtain  $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ .”

**Our formalization.** axiomatised

We take the cleaner truncated-distance bump  $f(x) = \min(\eta_3, d(x, x_0))$  in place of the paper’s hat. It is **machine-checked 1-Lipschitz** — `distBump_lipschitz`, obtained by truncating the 1-Lipschitz distance through the map  $t \mapsto \min(\eta_3, t)$  and routed through the `dist`-characterization to dodge the `EuclideanSpace/PiLp` extended-metric instance diamond — vanishes at  $x_0$ , and dominates  $\eta_3 \cdot \mathbf{1}_{\{d(\cdot, x_0) \geq \eta_3\}}$ . Hence  $\eta_3 \mu\{d(\cdot, x_0) \geq \eta_3\} \leq \int f d\mu = \int f d\mu - f(x_0) \leq W_1(\mu, \delta_{x_0}) \leq C\eta_2$ , and since  $\{\eta_3 \leq d(\cdot, x_0)\} = B(x_0, \eta_3)^c$  this is  $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$  for a probability measure. The Lipschitz bound is kernel-checked, but the middle inequality is Kantorovich-Rubinstein duality, supplied here by the labeled  $W_1$  axiom `W1_ge_of_lipschitz`; so the node as a whole is **axiomatised**.

```

/-- The truncated-distance bump  $f(x) = \min(\eta_3, d(x, x_0))$ . -/
noncomputable def distBump (x₀ : Eucl d) (η₃ : ℝ) (x : Eucl d) : ℝ := min η₃ (dist x x₀)

/-- The bump is '1'-Lipschitz (a truncation of the '1'-Lipschitz distance). Proved
  ↪ through the
'dist'-characterization to avoid the 'EuclideanSpace'/'PiLp' extended-metric instance
  ↪ diamond. -/
theorem distBump_lipschitz (x₀ : Eucl d) (η₃ : ℝ) : LipschitzWith 1 (distBump x₀ η₃) :=
  ↪ by
    rw [lipschitzWith_iff_dist_le_mul]
    intro x y
    -- 't ↦ min η₃ t' is 1-Lipschitz on ℝ (no PiLp instance diamond on the codomain)
    have hL : LipschitzWith 1 (fun t : ℝ => min η₃ t) := LipschitzWith.const_min
    ↪ LipschitzWith.id η₃
    have hmin := hL.dist_le_mul (dist x x₀) (dist y x₀)
    simp only [distBump, Real.dist_eq, NNReal.coe_one, one_mul] at hmin
    exact hmin.trans (abs_dist_sub_le x y x₀)

/-- **Lemma L8 / Claim 2 (Markov bound).** For a probability measure with
' $W_1(\mu, \delta_{\{x_0\}}) \leq C \eta_2$ ', the mass at geodesic/Euclidean distance  $\geq \eta_3$  from ' $x_0$ ' is at
  ↪ most
' $C \eta_2 / \eta_3$ '. Equivalently  $1 - \mu(B(x_0, \eta_3)) \leq C \eta_2 / \eta_3$ '. Rests on the ' $W_1$ ' axiom (KR
  ↪ duality):
status 'math.axiomatised'. -/
theorem markov_bound (μ : Measure (Eucl d)) [IsProbabilityMeasure μ]
  (x₀ : Eucl d) (η₂ η₃ C : ℝ) (hη₃ : 0 < η₃)
  (hW1 : W1 μ (Measure.dirac x₀) ≤ C * η₂) :
  μ.real {x | η₃ ≤ dist x x₀} ≤ C * η₂ / η₃ := by
  set s : Set (Eucl d) := {x | η₃ ≤ dist x x₀} with hs
  have hcont : Continuous (fun x : Eucl d => dist x x₀) := by fun_prop
  have hsmeas : MeasurableSet s := measurableSet_le measurable_const hcont.measurable
  -- the bump is integrable (bounded by η₃ on a finite measure)
  have hbump_int : Integrable (distBump x₀ η₃) μ := by
    refine Integrable.mono' (integrable_const η₃)
    (distBump_lipschitz x₀ η₃).continuous.aestronglyMeasurable ?_
    filter_upwards with x
    rw [Real.norm_eq_abs, abs_of_nonneg (distBump_nonneg hη₃.le x)]
    exact min_le_left _ _
  have hind_int : Integrable (s.indicator (fun _ => η₃)) μ :=
    (integrable_const η₃).indicator hsmeas
  -- η₃ · 1_s ≤ bump pointwise
  have hdom : s.indicator (fun _ => η₃) ≤ distBump x₀ η₃ := by
    intro x
    by_cases hx : x ∈ s
    · rw [Set.indicator_apply, if_pos hx]; exact (distBump_eq_outside hx).ge
    · rw [Set.indicator_apply, if_neg hx]; exact distBump_nonneg hη₃.le x
  --  $\int \eta_3 \cdot 1_s d\mu = \eta_3 \cdot \mu(s)$ 
  have hint_ind :  $\int x, s.indicator (fun _ => \eta_3) x \partial\mu = \mu.real s * \eta_3$  := by
    rw [integral_indicator_const η₃ hsmeas, smul_eq_mul]
  --  $\int \text{bump } d\delta_{\{x_0\}} = f(x_0) = 0$ 
  have hdirac :  $\int x, \text{distBump } x_0 \eta_3 x \partial(\text{Measure.dirac } x_0) = 0$  := by
    rw [integral_dirac]
    simp [distBump, min_eq_right hη₃.le]
  -- KR duality (axiom):  $\int \text{bump } d\mu - \int \text{bump } d\delta \leq W_1$ 
  have hdual := W1_ge_of_lipschitz μ (Measure.dirac x₀) (distBump x₀ η₃)
  ↪ (distBump_lipschitz x₀ η₃)
  rw [hdirac, sub_zero] at hdual
  -- assemble:  $\eta_3 \cdot \mu(s) \leq \int \text{bump } d\mu \leq W_1 \stackrel{20}{\leq} C \eta_2$ 
  have hmono :  $\mu.real s * \eta_3 \leq \int x, \text{distBump } x_0 \eta_3 x \partial\mu$  := by

```

### 3 Mass transport and clustering

The ReLU-gated drift  $\mathbf{W}(\mathbf{U}x+b)_+$  transports mass through a chain of overlapping balls, retaining a  $(1-\varepsilon)^K$  fraction of it after  $K$  parameter switches. Self-attention then clusters a measure supported in a single hemisphere to a point, and chaining the two steers an ensemble of disjoint-support atomless measures to prescribed discrete targets.

#### 3.1 Single overlapping ball pair (Lemma B.2)

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2, p. 31).

Consider two open balls  $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$  such that  $\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$ . For any  $\varepsilon > 0$  and  $T > 0$ , there exist  $\mathbf{W}, \mathbf{U} \in \mathcal{M}_{d \times d}(\mathbb{R})$  and  $b \in \mathbb{R}^d$  such that for any  $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$ , the unique solution  $\mu$  to (B.1) satisfies

$$\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1 - \varepsilon) \mu_0(\mathcal{B}_0). \quad (\text{B.3})$$

Moreover  $\mu(T) = \phi_{\#}^T \mu_0$  where the Lipschitz-continuous and invertible flow map  $\Phi^t : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  of (4.1) satisfies, for all  $t \in [0, T]$ ,

$$(\phi^t)|_{\mathbb{S}^{d-1} \setminus \mathcal{B}_0} \equiv \text{Id}.$$

Furthermore, for any fixed  $\omega \in \text{int } \mathcal{B}_0$  we can choose  $\mathbf{W}, \mathbf{U}$  and  $b$  so that the solution to (B.1) satisfies  $W_2(\mu(T), \alpha) \leq \varepsilon$  where  $\alpha(A) = \mu_0(\mathcal{B}_0) \delta_{\omega}(A) + \mu_0(A \setminus \mathcal{B}_0)$ , for any Borel  $A \subset \mathbb{S}^{d-1}$ .

*Proof sketch (full proof p. 32).* Let  $z$  and  $R$  be the center and radius of  $\mathcal{B}_0$  and fix  $\omega \in \text{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$ . The authors choose  $\mathbf{U}, b$  and a  $\mathbf{W}$  with  $\mathbf{W}\mathbf{1} = \omega$  so that the gated drift is  $(-\cos d_g(z, x) + \cos R)_+ \omega = (\cos R - \langle z, x \rangle)_+ \omega$ , and record (B.4)  $(\cos R - \langle z, x \rangle)_+ > 0 \iff x \in \mathcal{B}_0$  (see the strengthening below). Then (B.5)

$$\frac{d}{dt} \langle x(t), \omega \rangle = (\cos R - \langle z, x(t) \rangle)_+ (1 - \langle x(t), \omega \rangle^2),$$

which is positive whenever  $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$ . Hence for small  $\delta > 0$  there is a time  $T_{\delta}$  with  $\phi^{T_{\delta}}(x) \in \mathcal{B}_0 \cap \mathcal{B}_1$  for  $x \in B(z, R - \delta)$  (B.7); pushing forward,  $\mu(T_{\delta}, \mathcal{B}_0 \cap \mathcal{B}_1) = \mu_0((\phi^{T_{\delta}})^{-1}(\mathcal{B}_0 \cap \mathcal{B}_1)) \geq \mu_0(B(z, R - \delta))$  (B.8), and taking  $\delta$  small enough that  $\mu_0(B(z, R - \delta)) \geq (1 - \varepsilon) \mu_0(\mathcal{B}_0)$  gives (B.3) after rescaling time. The gate vanishes off  $\mathcal{B}_0$ , so  $\phi^t \equiv \text{Id}$  there. The bound to  $\alpha$  follows by taking  $\mathcal{B}_1 = B(\omega, \eta) \subset \mathcal{B}_0$  and a  $W_1$  Kantorovich-Rubinstein estimate, sending  $\eta$  and the slack to zero.

**What we strengthen.** As printed the proof takes  $\mathbf{U} = -\mathbf{1}z^{\top}$ ,  $b = \cos(R)\mathbf{1}$  (with  $\mathbf{W}\mathbf{1} = \omega$ ), so  $\mathbf{U}x = -\langle z, x \rangle \mathbf{1}$ , giving the gate coefficient  $(\cos R - \langle z, x \rangle)_+$ , and asserts in (B.4) that this is positive iff  $x \in \mathcal{B}_0 = B(z, R)$ . But

$$\cos R - \langle z, x \rangle > 0 \iff \langle z, x \rangle < \cos R \iff d_g(z, x) > R \iff x \notin \mathcal{B}_0,$$

so as written the gate is active on the **complement** of  $\mathcal{B}_0$ : the interior mass it is meant to move sees a zero gate, and the lemma transports nothing. The fix is a one-sign typo — take  $\mathbf{U} = +\mathbf{1}z^\top$ ,  $b = -\cos(R)\mathbf{1}$ , giving the gate  $(\langle z, x \rangle - \cos R)_+$ , positive exactly on  $\mathcal{B}_0$  — after which (B.3)-(B.5) and the rest of the argument go through and Lemma B.2 holds as stated. The corrected equivalence  $g(x) > 0 \iff d_g(z, x) > R$  is the kernel-checked leaf **L2** (`gate_pos_iff_dist`, [machine-checked](#)); the same inversion is corroborated independently by Proposition 4.2, Step 3 (which states the identical construction with the correct activation side) and by experiment E1. This is finding **F1**; see [ERRATA.md](#).

#### Our formalization. [open](#)

The authors’ strategy is the gated-drift construction above: a single ReLU gate, active on  $\mathcal{B}_0$  (after the F1 sign correction), pushes the interior mass of  $\mathcal{B}_0$  along  $+\omega$  into  $\mathcal{B}_0 \cap \mathcal{B}_1$  while leaving everything off  $\mathcal{B}_0$  fixed, so a  $(1 - \varepsilon)$  fraction is retained in the overlap. We state this type-correctly in Lean against the labeled axiom layer — the solution map `measureFlow`, the switch count `switches`, and the Wasserstein distance `W2` — with “no mass outside  $S$ ” read measure-theoretically. It is an open `sorry` stub: we do **not** claim a proof; discharging it rests on the continuity-equation flow axiom.

```
/-- **Lemma B.2** (single ball pair). Mass in ' $\mathcal{B}_0$ ' is pushed into ' $\mathcal{B}_0 \cap \mathcal{B}_1$ ', retaining
  ↪ ' $(1-\varepsilon)$ '. Stub
('sorry'): ' $\text{math.open}$ '. (Review finding F1: the paper's printed gate parameters have
  ↪ the activation
side reversed; the corrected sign is ' $\mathbf{U} = +\mathbf{1} z^\top$ ,  $b = -\cos(R) \mathbf{1}$ '. See 'ERRATA.md'.) -/
theorem lemma_B_2 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
  (B₀ B₁ : Set (Eucl d)) (hcap : (B₀ ∩ B₁).Nonempty) :
  ∃ θ : Params d, (1 - ENNReal.ofReal ε) * μ B₀ ≤ (measureFlow θ T μ) (B₀ ∩ B₁) := by
sorry
```

[Statements/MidLevel.lean](#)

### 3.2 Chain of overlapping balls (Lemma B.1)

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1, p. 31).

Consider  $K + 1$  open balls  $\mathcal{B}_K, \dots, \mathcal{B}_1, \mathcal{B}_0 \subset \mathbb{S}^{d-1}$  satisfying

$$\mathcal{B}_k \cap \mathcal{B}_{k-1} \neq \emptyset \quad \text{for } k \in \llbracket 1, K \rrbracket, \quad \mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset \quad \text{if } |k - k'| \geq 2.$$

Then for any  $T > 0$  and  $\varepsilon > 0$ , there exist  $(\mathbf{W}, \mathbf{U}, b) : [0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})^2 \times \mathbb{R}^d$ , piecewise constant having at most  $K$  switches, such that for any  $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$ , the corresponding unique solution  $\mu$  to

$$\begin{cases} \partial_t \mu(t) + \operatorname{div}(\mathbf{P}_x^\perp \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+ \mu(t)) = 0 & \text{on } [0, T] \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1}, \end{cases} \quad (\text{B.1})$$

satisfies

$$\mu(T, \mathcal{B}_K) \geq (1 - \varepsilon)^K \mu_0 \left( \bigcup_k \mathcal{B}_k \right).$$

Moreover,  $\mu(T) = \phi_{\#}^T \mu_0$  for a Lipschitz-continuous, invertible map  $\phi^t : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  which satisfies, for all  $t \in [0, T]$ ,

$$\phi^t(x) = x \quad \text{for } x \notin \bigcup_k \mathcal{B}_k. \quad (\text{B.2})$$

*Proof sketch (full proof p. 33).* Backward induction on the chain. Write  $[0, T] = \bigcup_{k \in \llbracket 1, K \rrbracket} [t_{k-1}, t_k]$  with  $t_k = kT/K$  and apply Lemma B.2 to the adjacent pair  $(\mathcal{B}_{k-1}, \mathcal{B}_k)$  on each subinterval. Then

$$\mu(T, \mathcal{B}_K) = \mu(T, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T, \mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1}, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1 - \varepsilon) \mu(t_{K-1}, \mathcal{B}_{K-1}),$$

the last step by Lemma B.2. The non-adjacency disjointness  $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$  for  $|k - k'| \geq 2$  keeps the steps from interfering, and iterating gives

$$\mu(T, \mathcal{B}_K) \geq (1 - \varepsilon)^K \left( \sum_{k=1}^K \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0) \right) = (1 - \varepsilon)^K \mu_0 \left( \bigcup_{k \in \llbracket 0, K \rrbracket} \mathcal{B}_k \right).$$

#### Our formalization. open

The authors' argument is a backward induction that applies the single-pair lemma (Lemma B.2) once per adjacent overlap, each step retaining a factor  $(1 - \varepsilon)$ , the non-adjacency disjointness guaranteeing the steps do not interfere. We state the retention bound type-correctly in Lean against the axiom layer (`measureFlow`, `switches`); it is an open `sorry` stub. The purely arithmetic heart — that a per-step retention factor  $(1 - \varepsilon)$  compounds to  $a_K \geq (1 - \varepsilon)^K a_0$  — is already discharged kernel-clean by leaf **L9** (`ball_chain_geom`, machine-checked); what remains open is the geometric non-interference, which rests on the parking property of the flow axiom.

```
-- **Lemma B.1** (ball-chain retention). For a chain of 'K+1' overlapping balls, 'K'
  ↳ switches retain
  '(1-ε)^K' of the mass into the last ball. Stub ('sorry'): 'math.open'. The arithmetic
  ↳ core is leaf
L9; the geometric non-interference is the parking axiom. -/
theorem lemma_B_1 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
  (K : ℕ) (B : ℕ → Set (Eucl d)) :
  ∃ θ : Params d, switches θ ≤ K ∧
    (1 - ENNReal.ofReal ε) ^ K * μ (⋃ k, B k) ≤ (measureFlow θ T μ) (B K) := by
  sorry
```

[Statements/MidLevel.lean](#)

### 3.3 Clustering a single-hemisphere measure to a point (Proposition 2.1)

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.1, p. 11).

Suppose  $\mathbf{B} \in \mathcal{M}_{d \times d}(\mathbb{R})$  and  $\text{supp } \mu_0$  is contained in an open hemisphere. Then the solution  $\mu$  to (1.3)-(1.2) with data  $\mu_0$  and  $(\mathbf{V}(\cdot), \mathbf{B}(\cdot), \mathbf{W}(\cdot)) \equiv (I_d, \mathbf{B}, 0)$  satisfies  $\text{diam}(\text{conv}_g \text{supp } \mu(t)) \rightarrow 0$

as  $t \rightarrow +\infty$ . Moreover, for  $\varepsilon > 0$  there exist  $z \in \text{conv}_g \text{supp } \mu_0$  and  $T > 0$  such that

$$W_\infty(\mu(T), \delta_z) \leq \varepsilon \quad \text{and} \quad \inf\{t \geq 0 : W_2(\mu(t), \delta_z) \leq \varepsilon\} = O\left(\log \frac{1}{\varepsilon}\right).$$

*Proof sketch (full proof p. 11; an adaptation of [GLPR25, Lemma 6.4]).* The characteristic flow is  $\dot{x}(t) = \mathbf{P}_{x(t)}^\perp \mathcal{A}_\mathbf{B}[\mu(t)](x(t))$ . Since  $\text{supp } \mu_0$  lies in an open hemisphere, the normalized attention vector  $\gamma(x) := \mathcal{A}_\mathbf{B}[\mu_0](x) / \|\mathcal{A}_\mathbf{B}[\mu_0](x)\|$  is well defined on  $\text{conv}_g \text{supp } \mu_0$  and points strictly into its interior. A first-order expansion at a boundary point  $x_0$  gives  $\langle x(\tau) - x_0, \gamma(x_0) \rangle > 0$  for small  $\tau > 0$ , so boundary points move inward and the geodesic convex hulls are strictly nested in time,  $\text{conv}_g \text{supp } \mu(t_2) \subset \text{int conv}_g \text{supp } \mu(t_1)$  for  $t_1 < t_2$ . Hence  $\phi(t) := \text{diam}(\text{conv}_g \text{supp } \mu(t))$  decreases to some  $\ell \geq 0$ ;  $\ell > 0$  would, by compactness, produce boundary points that fail to move inward, contradicting the strict interior-pointing, so  $\ell = 0$ . Nestedness and vanishing diameter yield a unique  $z$  with  $\mu \rightarrow \delta_z$ . Picking  $T$  with  $\text{diam}(\text{conv}_g \text{supp } \mu(T)) \leq \varepsilon$  and the coupling  $(\text{Id}, T)_\# \mu(T)$ ,  $T \equiv z$ , gives  $W_p(\mu(T), \delta_z) \leq \varepsilon$ ; the logarithmic rate follows from [CLPR25, Theorem 2.3].

#### Our formalization. open

The authors' strategy is a Lyapunov/LaSalle argument on the geodesic convex hull: with the pure self-attention drift ( $\mathbf{W} = 0$ ,  $\mathbf{V} = I_d$ ) the hull's diameter is strictly decreasing while its mass stays inside it, so the measure contracts to a single Dirac, the exponential contraction giving the  $O(\log \frac{1}{\varepsilon})$  time. We state this type-correctly in Lean — the open hemisphere is `supportedIn μ {x | 0 < ⟨e, x⟩}` for a unit  $e$ , the target a `Measure.dirac z`, closeness measured in  $W_2$  — as an open `sorry` stub. We do **not** claim a proof: the diameter monotonicity, the LaSalle/Hartman-Grobman convergence, and the rate live in the continuity-equation flow axiom. The conclusion is validated numerically by experiment **E2**.

```
-- **Proposition 2.1** (clustering to a point). A measure supported in an open
  ↪ hemisphere can be
driven arbitrarily 'W2'-close to a Dirac mass. Stub ('sorry'): 'math.open'. The rate
'O(log 1/ε)' and the LaSalle/Hartman-Grobman convergence are in the axiom layer. -/
theorem prop_2_1 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
  (e : Eucl d) (he : ‖e‖ = 1) (hhemi : supportedIn μ {x | 0 < ⟨e, x⟩}) :
  ∃ (θ : Params d) (z : Eucl d), W2 (measureFlow θ T μ) (Measure.dirac z) ≤ ε := by
sorry
```

[Statements/MidLevel.lean](#)

### 3.4 Clustering disjoint-support measures to atoms (Proposition 2.2)

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.2, pp. 11-12). Suppose  $\mu_0^i$  has no atoms and  $\text{conv}_g \text{supp } \mu_0^i \cap \text{conv}_g \text{supp } \mu_0^j = \emptyset$  for  $i \neq j$ . Fix  $M \geq 1$ , and for any  $i \in \llbracket 1, N \rrbracket$  consider

$$\mu_1^i := \sum_{k=1}^M \alpha_k^i \delta_{x_k^i} \in \mathcal{P}(\mathbb{S}^{d-1})$$



where  $x_k^i \in \text{conv}_g \text{supp } \mu_0^i$ , with  $x_k^i = x_{k'}^j$  if and only if  $(k, i) = (k', j)$ . Then for any  $T > 0$  and  $\varepsilon > 0$  there exist piecewise constant  $(\mathbf{W}, \mathbf{U}, b) : [0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})^2 \times \mathbb{R}^d$  such that for any  $i$ , the corresponding solution  $\mu^i$  to (1.3)-(1.2) with data  $\mu_0^i$ ,  $\mathbf{V} \equiv 0$  and the above parameters, satisfies  $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$  as well as, for  $i \neq j$ ,

$$\text{conv}_g \text{supp } \mu^i(T) \cap \text{conv}_g \text{supp } \mu^j(T) = \emptyset.$$

The number of switches in  $(\mathbf{W}, \mathbf{V}, b)$  can also be accounted for — see Remark 2.3.

*Proof sketch (full proof from p. 12).* Split  $[0, T] = \bigcup_{i \in [1, N]} [T_{i-1}, T_i]$  and apply Lemma B.1 separately within each interval, acting on one measure at a time. Property (B.2) — the flow is the identity off the active balls — ensures that while the construction acts on the  $i$ -th measure the others are untouched,  $\mu^i(T_{i-1}) = \mu_0^i$ . For a fixed  $i$ , partition the support  $\mathcal{C}^i = \text{supp } \mu_0^i$  into  $M$  pairwise disjoint pieces  $\mathcal{C}_k^i$  with connected interiors,  $\mu_0^i(\mathcal{C}_k^i) = \alpha_k^i$  and  $x_k^i \in \text{int } \mathcal{C}_k^i$ ; then steer the mass of each piece, through a chain of overlapping balls, into a small ball around  $x_k^i$ , so  $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$  while the disjoint supports remain disjoint.

### Our formalization. open

The authors' strategy chains the two preceding results: partition each atomless support into  $M$  prescribed-mass pieces, then use the ball-chain transport (Lemma B.1) to drive each piece into a small ball about its target atom  $x_k^i$ , handling the disjoint families one at a time so that — by the off-support identity (B.2) — the others stay fixed. We state the single-family conclusion type-correctly in Lean against the axiom layer: the target is a genuine  $M$ -atom combination  $\sum k, \alpha k \bullet \text{Measure.dirac } (x k)$ , atomlessness is  $\forall x, \mu \{x\} = 0$ , and closeness is in  $W_2$ . It is an open sorry stub — not a claimed proof — resting on the flow and optimal-transport axioms together with the (open) Lemma B.1 and Proposition 2.1.

```
/-- **Proposition 2.2** (clustering to a discrete measure). An atomless measure whose
  ↳ support lies in
  a region 'S' disjoint from the others can be driven 'W2'-close to a prescribed 'M'-atom
  ↳ empirical
measure. Stub ('sorry'): 'math.open'. -/
theorem prop_2_2 (μ v_target : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
  (hatomless : ∀ x : Eucl d, μ {x} = 0)
  (M : ℕ) (x : Fin M → Eucl d) (α : Fin M → ℝ≥0∞)
  (htgt : v_target = ∑ k : Fin M, α k • Measure.dirac (x k)) :
  ∃ θ : Params d, W2 (measureFlow θ T μ) v_target ≤ ε := by
  sorry
```

[Statements/MidLevel.lean](#)

## 4 Disentangling supports

A piecewise-constant control separates the  $N$  overlapping input supports into pairwise-disjoint regions of the sphere, by steering each measure's *barycenter* into its own direction and then shrinking its geodesic hull around it. This disentanglement is the heart of the construction behind Theorems 1.1 and 1.2: once the supports are disjoint, each measure can be clustered and matched

independently.

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, §3, p. 14).

“We show that flows generated by self-attention can disentangle measures with overlapping supports—it actually suffices to consider  $\mathbf{B} \equiv 0$ .” Set  $\mathbb{Q}_1^{d-1} := \mathbb{S}^{d-1} \cap (\mathbb{R}_{>0})^d$  and

$$\mathbf{v}[\mu](t, x) = \mathbf{P}_x^\perp(\mathbf{V}(t) \mathbb{E}_{\mu(t)}[z] + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+). \quad (3.1)$$

The whole section uses this reduced field (3.1): a perceptron term  $\mathbf{W}(\mathbf{U}x + b)_+$  and a barycentric attention term  $\mathbf{V} \mathbb{E}_\mu[z]$ , both projected onto the tangent space.

## 4.1 Transport to the orthant

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.2, p. 15).

Suppose  $T > 0$  and  $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$  with  $\bigcup_i \text{supp } \mu_0^i \subsetneq \mathbb{S}^{d-1}$ . There exists a piecewise constant  $\mathbf{W} : [0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})$ , having at most one switch and satisfying

$$\|\mathbf{W}\|_{L^\infty((0,T);\mathcal{M}_{d \times d}(\mathbb{R}))} \leq C/T$$

for some  $C = C(N) > 0$ , such that for any  $i \in \llbracket 1, N \rrbracket$  the solution  $\mu^i$  to (1.3)–(1.2) with data  $\mu_0^i$  and  $\mathbf{V} \equiv \mathbf{B} \equiv \mathbf{U} \equiv 0$ ,  $b \equiv \mathbf{1}$ , satisfies  $\text{supp } \mu^i(T) \subset \mathbb{Q}_1^{d-1}$ .

**Proof.** (*proof sketch; full proof p. 15.*) Pick a direction  $\omega \in \mathbb{S}^{d-1} \setminus \bigcup_i \text{supp } \mu_0^i$  not filling the sphere and set  $\mathbf{W}_1 \mathbf{1} = -\omega$ . The characteristics solve  $\dot{x} = -\mathbf{P}_x^\perp \omega$ , so  $\frac{d}{dt} \langle x, \omega \rangle = -1 + \langle x, \omega \rangle^2 < 0$  off  $\{\pm\omega\}$ , hence by some  $T_0$  all supports lie in  $B(-\omega, \pi/8)$ . Pick  $\alpha \in \mathbb{Q}_1^{d-1}$  with  $d_g(-\omega, -\alpha) > \pi/8$  (so  $-\alpha$  is off every support) and on  $[T_0, T]$  take  $\mathbf{W}_2 \mathbf{1} = \alpha$ : now  $\frac{d}{dt} \langle x, \alpha \rangle = 1 - \langle x, \alpha \rangle^2 \geq 0$  drives every point into a prescribed small cap around  $\alpha \in \mathbb{Q}_1^{d-1}$ . One switch; the norm bound follows by time-rescaling the two phases.  $\square$

### Our formalization. open

The authors’ construction (attributed to them) is the one switch above: a first phase  $\mathbf{W}_1 \mathbf{1} = -\omega$  contracts every support toward  $-\omega$  through the gate-free dynamics  $\frac{d}{dt} \langle x, \omega \rangle = -(1 - \langle x, \omega \rangle^2)$  — the projector identity of leaf **L1** — and a second phase  $\mathbf{W}_2 \mathbf{1} = \alpha \in \mathbb{Q}_1^{d-1}$  drives them into a small cap inside the orthant. We give a **type-correct Lean statement against the axiom layer** (`measureFlow`, `switches`), expressing “support inside  $\mathbb{Q}_1^{d-1}$ ” measure-theoretically as  $\mu \text{ (orthant d)}^c = 0$ . It is a **sorry stub**: we state it, we do **not** claim a proof; its discharge rests on the continuity-equation flow axiom.

```

/-- The open positive orthant  $Q_1^{d-1} = S^{d-1} \cap (\mathbb{R}_{>0})^d$ , as a subset of  $\mathbb{R}^d$ . -/
def orthant (d : ℕ) : Set (Eucl d) := {x | ∀ i, 0 < x i}

/-- "The support of  $\mu$  is contained in  $S$ ", expressed measure-theoretically as  $\mu(S^c) \hookrightarrow = 0$ . -/
def supportedIn (μ : Measure (Eucl d)) (S : Set (Eucl d)) : Prop := μ Sc = 0

/-- **Lemma 3.2** (transport into the orthant). One parameter switch moves the measure
    into
 $Q_1^{d-1}$ . Stub ('sorry'): 'math.open'. -/
theorem lemma_3_2 (μ : Measure (Eucl d)) (T : ℝ) (hT : 0 < T) :
  ∃ θ : Params d, switches θ ≤ 1 ∧ supportedIn (measureFlow θ T μ) (orthant d) := by
  sorry

```

[Statements/MidLevel.lean](#)

## 4.2 Shrinking a hull to its barycenter direction

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.3, p. 16; proof App. B.2, p. 33).

Let  $\mu_0^i \in \mathcal{P}(\mathbb{Q}_1^{d-1})$  be such that  $\mathbb{E}_{\mu_0^i}[x]$  is not colinear with  $\mathbb{E}_{\mu_0^j}[x]$  for  $i \neq j$ . Then for any  $T > 0$ ,  $\varepsilon > 0$ , and  $\nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$  such that  $\mathbb{E}_{\nu_0}[x]$  is colinear with  $\mathbb{E}_{\mu_0^i}[x]$ , there exists a piecewise constant  $\theta : [0, T] \rightarrow \Theta$  having at most  $O(d \cdot N)$  switches such that

$$\text{conv}_g \text{supp } \nu(T) \cup \text{conv}_g \text{supp } \mu^N(T) \subset B\left(\mathbb{E}_{\mu_0^j}[z] / \|\mathbb{E}_{\mu_0^j}[z]\|, \varepsilon\right)$$

and  $\mu^i(T) = \mu_0^i$  for  $i \neq j$ , where  $\mu^i, \nu$  denote the unique solutions to (1.3)–(3.1) corresponding to data  $\mu_0^i, \nu_0$ , and parameters  $\theta$ .

**Proof.** (*proof sketch; full proof App. B.2, p. 33.*) Three steps. **(1) Isolating**  $\mu_0^N, \nu_0$ : with  $\mathbf{W} \equiv 0$  and  $\mathbf{V}(t) = \sum_k \alpha_k \alpha_k^\top \mathbf{1}_{[T_{k-1}, T_k]}$ , where  $\{\alpha_k\}$  is an orthonormal basis of  $\text{span}(\mathbb{E}_{\mu_0^N}[z])^\perp$ , the barycenter ODE (B.9)  $\frac{d}{dt} \langle x, \alpha_1 \rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1 \rangle (1 - \langle \alpha_1, x \rangle^2)$  sends  $x(t) \rightarrow \pm \alpha_k$ , concentrating  $\text{supp } \mu(T_{d-1})$  near  $\bigcup_k B(\pm \alpha_k, C_k \varepsilon_k) \subset \mathbb{S}^{d-1} \setminus \mathbb{Q}_1^{d-1}$  (flow  $\Psi_1$ , fixing  $\mu_0^j$  for  $j \leq N-1$ ). **(2) Clustering**: a separating half-space  $\langle a, x \rangle + \underline{b}$  with  $\mathbf{U} = \mathbf{1} a^\top$ ,  $\mathbf{W}_2 \mathbf{1} = \mathbb{E}_{\mu_0^N}[x]$  drives both supports into  $B(\mathbb{E}_{\mu_0^N}[x] / \|\mathbb{E}_{\mu_0^N}[x]\|, \delta)$  (flow  $\Psi_2$ ). **(3) Flow reversal**:  $\Phi_{\text{fin}} := \Psi_1^{-1} \circ \Psi_2 \circ \Psi_1$  returns  $\mu_0^i$  to itself for  $i \leq N-1$  and clusters  $\nu, \mu^N$  within  $B(\cdot, C_T \delta)$ ; pick  $\delta$  small.  $\square$

### Our formalization. open

The authors' three-step argument (App. B.2) hinges on the **barycenter ODE (B.9)**, which is exactly our machine-checked leaf **L6** (`barycenter_hasDerivAt_inner`): along the attention field,  $\frac{d}{dt} \langle x, \alpha \rangle = c(1 - \langle x, \alpha \rangle^2)$  with strict increase for  $c > 0$ , which is what drives each point to  $\pm \alpha$  and lets the support be concentrated into an  $\varepsilon$ -ball around the barycenter direction. We state the conclusion type-correctly against the axiom layer: some  $\theta$  concentrates the flow into `Metric.ball`  $\alpha$   $\varepsilon$ . It is a `sorry` stub stated against `measureFlow`; the analytic content (the three flows  $\Psi_1, \Psi_2, \Psi_1^{-1}$ ) lives in the flow axiom.

```

/-- **Lemma 3.3** (shrink a measure's hull toward its barycenter direction). For any
  ↪ tolerance the
measure can be concentrated into a small ball around some direction `α`, with `O(dN)`
  ↪ switches.
Stub (`sorry`): `math.open`. -/
theorem lemma_3_3 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε) :
  ∃ (θ : Params d) (α : Eucl d), supportedIn (measureFlow θ T μ) (Metric.ball α ε) :=
  ↪ by
  sorry

```

Statements/MidLevel.lean

### 4.3 Making barycenters non-colinear

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4, p. 16; proof App. B.3, p. 35).

Let  $T > 0$  and let  $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$  be two different measures such that  $\mathbb{E}_{\mu_0}[x] = \gamma_1 \mathbb{E}_{\nu_0}[x]$  for some  $\gamma_1 \in (0, 1]$ .

1. If  $\gamma_1 = 1$ , then, setting  $\mathbf{V} \equiv 0$ , there exist  $\mathbf{W}, \mathbf{U} \in \mathcal{M}_{d \times d}(\mathbb{R})$  and  $b \in \mathbb{R}^d$  such that the solutions  $\mu, \nu$  to (1.3)–(3.1) corresponding to  $\mu_0, \nu_0$  and these parameters, satisfy  $\mathbb{E}_{\mu(T)}[x] \neq \mathbb{E}_{\nu(T)}[x]$ . Moreover the Lipschitz-continuous and invertible flow map  $\Phi^T : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  induced by the characteristics satisfies  $\Phi^T(x) = x$  for  $x \in \mathbb{S}^{d-1} \setminus (\text{conv}_g \text{supp } \mu_0 \cup \text{conv}_g \text{supp } \nu_0)$ . (3.2)
2. If  $\gamma_1 \neq 1$ , then, setting  $\mathbf{B} \equiv 0$ , there exist piecewise-constant  $(\mathbf{V}, \mathbf{W}, \mathbf{U})$  and  $b$  with at most 2 switches such that  $\mathbb{E}_{\mu(T)}[x] \neq \gamma_2 \mathbb{E}_{\nu(T)}[x]$  for all  $\gamma_2 \in \mathbb{R}$  (non-colinear).

**Proof (Part 1).** (*proof sketch; full proof App. B.3, p. 35.*) Since  $\mu_0 \neq \nu_0$ , there is an open ball  $\mathcal{B} \subset \text{supp } \mu_0 \cup \text{supp } \nu_0$  with  $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$ . There exists  $x^* \in \mathcal{B}$  with

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x \mu_0(dx) \neq \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x \nu_0(dx),$$

for otherwise  $x^* = \frac{1}{\mu_0(\mathcal{B}) - \nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1} \setminus \mathcal{B}} x (\nu_0 - \mu_0)(dx)$  would be forced constant over all of  $\mathcal{B}$  — impossible. Taking such  $x^*$ , a perceptron block  $\mathbf{U} = -\mathbf{1}a^\top$ ,  $b = R\mathbf{1}$  ( $a, R$  the center and radius of  $\mathcal{B}$ ), and  $\mathbf{W}\mathbf{1} = x^*$ , by Lemma B.2 pushes  $\mu(T)$  to  $\mu_0(\mathcal{B})\delta_{x^*} + \mu_0(\cdot \setminus \mathcal{B})$ ; continuity of the expectation in  $W_2$  yields  $\mathbb{E}_{\Phi_{\#}\mu_0}[x] \neq \mathbb{E}_{\Phi_{\#}\nu_0}[x]$ , with  $\Phi(x) = x$  off  $\mathcal{B}$ .  $\square$

#### Our formalization. open

In the  $\gamma_1 = 1$  (perceptron) case the authors locate a ball  $\mathcal{B}$  on which  $\mu_0, \nu_0$  differ in mass, then argue that **some**  $x^* \in \mathcal{B}$  **makes the barycenters differ** — for if every  $x^*$  gave equality,  $x^*$  would be pinned to one value on the whole ball, which no map can be. That “no map is constant on a nonempty open ball” core is our machine-checked leaf **L10** (`exists_ne_in_ball`). We state the  $\gamma_1 = 1$  conclusion type-correctly: equal input barycenters force the existence of

a  $\theta$  whose output barycenters differ. `sorry` stub against the axiom layer. (The attention case  $\gamma_1 \neq 1$ , at most two switches, is the companion stub `lemma_3_4_part2`.)

```

/-- **Lemma 3.4, Part 1** (' $\gamma_1 = 1$ ' case). If two measures have equal barycenters, a
  ↪ constant
parameter makes the barycenters differ. Stub ('sorry'): ' $\mathsf{math.open}$ '. The self-contained
  ↪ pigeonhole
core is leaf L10. -/
theorem lemma_3_4_part1 (μ ν : Measure (Eucl d)) (T : ℝ) (hT : 0 < T)
  (hbar : barycenter μ = barycenter ν) :
  ∃ θ : Params d, barycenter (measureFlow θ T μ) ≠ barycenter (measureFlow θ T ν) := by
  sorry

```

[Statements/MidLevel.lean](#)

## 4.4 Disentanglement

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1, pp. 15-17).

Let  $T > 0$  and  $\mu_0^i \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ . There exists a piecewise constant  $\theta : [0, T] \rightarrow \Theta$ , having at most  $O(d \cdot N)$  switches, such that for all  $i \in \llbracket 1, N \rrbracket$ , the solution  $\mu^i$  to (1.3)–(3.1) with data  $\mu_0^i$  and  $\theta$  satisfies

$$\text{conv}_g \text{supp } \mu^i(T) \cap \text{conv}_g \text{supp } \mu^j(T) = \emptyset \quad \text{if } i \neq j.$$

**Proof.** (*proof sketch; full proof pp. 16-17.*) Induction on  $N$ ; the base case  $N = 1$  is trivial. Assume the first  $N - 1$  measures already have pairwise-disjoint geodesic hulls (3.3) and let  $\mu_0^N$  be arbitrary. Since  $\text{supp } \mu_0^i \subset \mathbb{Q}_1^{d-1}$ , (3.3) implies  $\mathbb{E}_{\mu_0^i}[x]$  is not colinear with  $\mathbb{E}_{\mu_0^j}[x]$  for  $i \neq j \leq N - 1$ . If  $\mathbb{E}_{\mu_0^N}[x]$  is non-colinear with all of them, one application of Lemma 3.3 (small  $\varepsilon$ ) shrinks  $\mu^N$ 's hull until it separates. Otherwise  $\mathbb{E}_{\mu_0^N}[x]$  is colinear with  $\mathbb{E}_{\mu_0^i}[x]$  for at most one  $i$  (WLOG  $i = N - 1$ ), and four phases of  $[0, T]$  finish it: (1)  $[0, T/4]$  Lemma 3.3 peels  $\mu^N, \mu^{N-1}$  off the rest; (2)  $[T/4, T/2]$  Lemma 3.4 Part 1 makes  $\mathbb{E}_{\mu^{N-1}}[x] \neq \mathbb{E}_{\mu^N}[x]$  while (3.2) keeps the others fixed; (3)  $[T/2, 3T/4]$  Lemma 3.4 Part 2 makes the two barycenters non-colinear; (4)  $[3T/4, T]$  Lemma 3.3, now applicable to all  $N$  measures, separates them. Total  $O(d \cdot N)$  switches.  $\square$

### Our formalization. open

We state the headline controllability conclusion of the authors' induction type-correctly against the axiom layer: there is a  $\theta$  whose solution map renders the family's supports pairwise disjoint, recorded by the opaque predicate `DisjointSupports`. It is a `sorry` stub — the four-phase induction, the  $O(dN)$  switch count, and the geodesic-hull nesting all rest on the flow /  $\text{conv}_g$  axioms, so we do not claim a proof.

```

/-- The measures in the family have pairwise disjoint (geodesic-convex-hull) supports.
↪ -/
axiom DisjointSupports {N : ℕ} (ν : Fin N → Measure (Eucl d)) : Prop

/-- **Proposition 3.1** (disentanglement). There is a piecewise-constant `θ` whose
↪ solution map
renders the supports of the family pairwise disjoint. Stub (`sorry`): status
↪ `math.open`. -/
theorem prop_3_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (T : ℝ) (hT : 0 <
↪ T) :
  ∃ θ : Params d, DisjointSupports (fun i => measureFlow θ T (μ₀ i)) := by
  sorry

```

[Statements/MainResults.lean](#)

**What we strengthen.** (*finding F2.*) The induction above opens by asserting that **disjoint geodesic hulls imply non-colinear barycenters** —

$$\text{conv}_g \text{supp } \mu_0^i \cap \text{conv}_g \text{supp } \mu_0^j = \emptyset \implies \mathbb{E}_{\mu_0^i}[x] \text{ not colinear with } \mathbb{E}_{\mu_0^j}[x]$$

— **without proof.** Our adversarial review flagged this as the one genuine rigor gap (finding F2). It is now the machine-checked leaf **L11** (`barycenter_noncolinear_of_disjoint_hull`, see the [Self-contained leaves](#) section) for the empirical regime of Theorem 1.1 / restricted Theorem 1.2, and **L11'** (`barycenter_noncolinear_of_disjoint_hull_general`) for arbitrary probability measures via Mathlib’s `Convex.integral_mem`. The barycenter, a nonnegative average of support points, lies in  $\text{cone}(\text{supp})$ ; colinearity would put a common normalized direction in *both* hulls, contradicting disjointness. The headline `prop_3_1` stays `[open]{.badge .open}` only because it still rests on the flow /  $\text{conv}_g$ -nesting axioms — not on this step. (Validated numerically by experiment E3.)

[Leaves/BarycenterNonColinear.lean](#)

## 5 Matching discrete measures

Once the disentanglement of §3 has rendered the supports pairwise disjoint and the clustering of §2 has collapsed each measure to a single point, matching reduces to steering finitely many points to their prescribed targets. A single perceptron gate — identically zero on a chosen parking cap and positive outside it — lets the flow drive one *active* point along a geodesic corridor to its target while every other point, parked inside the cap, stays fixed.

## 5.1 Proposition 4.2 — steer the active point

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.2, p. 18). Suppose  $d \geq 3$ . Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \quad \text{for } i \in \llbracket 1, M \rrbracket, \quad (\mathcal{D})$$

with  $x_0^i \neq x_0^j$  and  $y^i \neq y^j$  for  $i \neq j$ , with  $x_0^i = y^i$  for  $i \in \llbracket 1, M-1 \rrbracket$ , and suppose that there exist  $\gamma \in \mathbb{S}^{d-1}$  and  $\varepsilon > 0$  such that  $\langle \gamma, x_0^M - y^M \rangle = 0$  and  $x_0^i \notin H_\varepsilon^\gamma$  for all  $i \in \llbracket 1, M-1 \rrbracket$ , where  $H_\varepsilon^\gamma := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma \rangle| \leq \varepsilon\}$ .

For any  $T > 0$  there exist piecewise constant  $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})^2 \times \mathbb{R}^d$ , having at most 6 switches, such that for any  $i$ , the solution  $x^i(\cdot)$  to

$$\begin{cases} \dot{x}^i(t) = \mathbf{P}_x^\perp \mathbf{W}(t)(\mathbf{U}(t)x^i(t) + b(t))_+ & \text{in } [0, T] \\ x^i(0) = x_0^i, \end{cases} \quad (4.1)$$

satisfies  $x^i(T) = y^i$ . Moreover, there exists  $C > 0$ , not depending on  $\mathcal{D}$  and  $T$ , such that

$$\|\theta\|_{L^\infty((0,T);\Theta)} \leq \frac{C}{T \cdot \varepsilon}.$$

**Proof sketch** (proof sketch; full proof pp. 18-23). The control is a 6-interval step function  $(\mathbf{W}(t), \mathbf{U}(t), b(t)) = \sum_{j=1}^6 (\mathbf{W}_j, \mathbf{U}_j, b_j) \mathbf{1}_{[(j-1)T/6, jT/6]}(t)$  with  $\mathbf{U}_5 = \mathbf{U}_1$ ,  $\mathbf{U}_6 = \mathbf{U}_2$ ,  $\mathbf{U}_3 = \mathbf{U}_4$ ,  $b_5 = b_1$ ,  $b_6 = b_2$ ,  $b_3 = b_4$ ,  $\mathbf{W}_5 = -\mathbf{W}_1$ ,  $\mathbf{W}_6 = -\mathbf{W}_2$ , i.e. the last two intervals time-reverse the first two — at most 6 switches. *Anchors (Step 1)*. Since  $\langle \gamma, x_0^M - y^M \rangle = 0$  one finds  $\omega \in \mathbb{S}^{d-1}$  with  $\langle \gamma, \omega \rangle = 0$  and  $d_g(\omega, x_0^M), d_g(\omega, y^M) \geq \pi/2$ ; the points  $\omega_\pm$  at geodesic distance  $\pi/8$  from  $\omega$  toward  $\pm\gamma$  then satisfy  $d_g(\omega_\pm, x_0^M), d_g(\omega_\pm, y^M) \geq 3\pi/8$ , so  $\{\langle \omega, x \rangle = \cos(\pi/8 + \tau)\}$  separates the cap  $B(\omega, \pi/8 + \tau)$  from  $x_0^M, y^M$ . *Phase 1 — “gather”  $\psi_1$  (Steps 1-2, 2 switches)*. Two opposite gates  $(\langle \pm\gamma, x \rangle - \varepsilon/2)_+$  with drift targets  $\omega_\pm$  run the Riemannian gradient flow of  $d_g(\cdot, \omega_\pm)$  (using  $\nabla_1 d_g(x, \omega_+) = -\mathbf{P}_x^\perp \omega_+ / \sqrt{1 - \langle x, \omega_+ \rangle^2}$ );  $\langle x, \omega_\pm \rangle$  increases monotonically (4.5) and the trajectory settles exponentially near  $\omega_\pm$  by LaSalle / Hartman-Grobman (4.6), so after time  $T/3$  every inactive point lies in the cap  $B(\omega, 3\pi/16)$  while points of  $H_\varepsilon^\gamma$  stay fixed (4.7). *Phase 2 — “corridor”  $\psi_2$  (Step 3, 2 switches)*. The gate  $(\langle -\omega, x \rangle + \cos(3\pi/16))_+$  is identically 0 on  $B(\omega, 3\pi/16)$  and positive outside; driving  $x_0^M$  first toward an intermediate  $z_1$  then toward  $z_2$  along a geodesic corridor that lies outside the cap and passes through  $y^M$  lands the active point at  $y^M$ , while the inactive points do not move (gate off on the cap). *Phase 3 — “restore”  $\psi_1^{-1}$  (Step 4, 2 switches)*. Re-running the gather action with flipped drifts time-reverses Phase 1, returning every inactive point to its start; the active point remains at  $y^M$  since the corridor gate is off on the cap. The composition  $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$  sends  $x_0^M \mapsto y^M$  and fixes all other  $x_0^i$ ; choosing drift size  $\|\mathbf{W}\| \asymp (T\varepsilon)^{-1}$  gives total time  $T$  and  $\|\theta\|_{L^\infty} \lesssim 1/(T\varepsilon)$ .

### Our formalization. open

The authors’ construction (pp. 18-23) realises the matching of one point as the gather / corridor / restore composition  $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$ , a 6-interval piecewise-constant control whose middle phase steers the active  $x_0^M$  to  $y^M$  while a gate parked on the cap  $B(\omega, 3\pi/16)$  freezes the inactive points. We record a type-correct Lean **statement** of this proposition as a `sorry`



(open): with  $d \geq 3$ , the inactive points already at their targets (hfix), at most 6 switches drive every input to its target while keeping the inactive ones fixed. We do **not** claim a proof. Two of the construction's cores are however discharged as kernel-checked leaves: the separating-hyperplane estimate of Step 1 is leaf **L3** ([Leaves/SeparatingHyperplane.lean](#), `separating_hyperplane`), and the geodesic-gradient identity driving the gather and corridor flows is leaf **L4** ([Leaves/GeodesicGradient.lean](#), `geodesicDist_hasDerivAt`). The selective-motion mechanism — active point reaches the target, parked points stay fixed — is validated numerically by experiment **E4**.

```

/-- **Proposition 4.2** (steer one active point). With `d ≥ 3` and the inactive points
  ↪ already at
their targets, at most `6` switches move every input to its target, keeping the inactive
  ↪ ones fixed
(`x₀ i = y i` is preserved by `flowMap θ T (x₀ i) = y i`). Stub (`sorry`): `math.open`.
  ↪ Step 1 is
leaf L3, the geodesic gradient is leaf L4. -/
theorem prop_4_2 (hd : 3 ≤ d) (M : ℕ) (x₀ y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)
  (hfix : ∀ i : Fin M, (i : ℕ) < M - 1 → x₀ i = y i) :
  ∃ θ : Params d, switches θ ≤ 6 ∧ ∀ i, flowMap θ T (x₀ i) = y i := by
  sorry

```

[Statements/MidLevel.lean](#)

## 5.2 Proposition 4.1 — match the ensemble

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.1, p. 18). Suppose  $d \geq 3$ . Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \quad \text{for } i \in \llbracket 1, M \rrbracket, \quad (\mathcal{D})$$

with  $x_0^i \neq x_0^j$  and  $y^i \neq y^j$  for  $i \neq j$ , and suppose that for any  $i$  there exist  $\gamma_i \in \mathbb{S}^{d-1}$  and  $\varepsilon_i > 0$  such that  $\langle \gamma_i, x_0^i - y^i \rangle = 0$  and  $x_0^j \notin H_{\varepsilon_i}^{\gamma_i}$  for  $j \neq i$ , where

$$H_{\varepsilon_i}^{\gamma_i} := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma_i \rangle| \leq \varepsilon_i\}.$$

For any  $T > 0$  there exist piecewise constant  $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \rightarrow \mathcal{M}_{d \times d}(\mathbb{R})^2 \times \mathbb{R}^d$ , having at most  $6M$  switches, such that for any  $i$ , the solution  $x^i(\cdot) \in \mathcal{C}^0([0, T]; \mathbb{S}^{d-1})$  to (4.1) satisfies  $x^i(T) = y^i$ . Moreover, there exists  $C > 0$ , not depending on  $\mathcal{D}$  nor  $T$ , such that

$$\|\theta\|_{L^\infty((0, T); \Theta)} \leq \frac{C \cdot M}{T \min_i \varepsilon_i}.$$

**Proof sketch** (proof sketch; full proof p. 18). “The proof of Proposition 4.1 follows directly from [Proposition 4.2], combined with a straightforward induction argument”: activating one pair  $(x_0^i, y^i)$  at a time, each application of Proposition 4.2 steers the  $i$ -th point to its target in at most 6 switches while leaving the already-matched points fixed, so  $M$  applications cost at most  $6M$  switches and the norm bound accumulates to  $C \cdot M / (T \min_i \varepsilon_i)$ .



### Our formalization. open

The authors obtain Proposition 4.1 by induction over the  $M$  pairs, each step being one application of Proposition 4.2 (the active point reaches its target; the previously matched points are inactive and stay fixed), giving the  $\leq 6M$ -switch budget and the  $O(CM/(T \min_i \varepsilon_i))$  norm. We record a type-correct Lean **statement** as a **sorry** (open): with  $d \geq 3$  and injective inputs/targets, at most  $6M$  switches steer every  $x_0^i$  to  $y^i$ . We do **not** claim a proof; the discharge would be the  $M$ -fold composition of **prop\_4\_2**. (Validated numerically by experiment **E4**, alongside Proposition 4.2.)

```
/-- **Proposition 4.1** (match an ensemble). With `d ≥ 3` and distinct inputs/targets,  
  ↪ at most `6M`  
switches steer every `x_0 i` to `y i`. Stub (`sorry`): `math.open`. Follows from  
  ↪ Proposition 4.2 by  
induction. -/  
theorem prop_4_1 (hd : 3 ≤ d) (M : ℕ) (x_0 y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)  
  (hx_0 : Function.Injective x_0) (hy : Function.Injective y) :  
  ∃ θ : Params d, switches θ ≤ 6 * M ∧ ∀ i, flowMap θ T (x_0 i) = y i := by  
  sorry
```

[Statements/MidLevel.lean](#)

## 6 Main results

The paper assembles its leaves and mid-level propositions into one solution map  $\Phi_{\text{fin}}^T$  by a three-stage recipe: **disentangle** the  $N$  supports until they are pairwise disjoint (Proposition 3.1), **cluster** each measure — to a single point (Proposition 2.1) or to an  $M$ -atom empirical measure (Proposition 2.2) — and **match** each cluster to its prescribed target (Propositions 4.1/4.2). Theorem 1.1 is the Dirac-target case, steering each input to a point mass  $\delta_{x^i}$  with  $O(dN)$  switches and an explicit norm bound, while Theorem 1.2 is the general case, reaching any  $L^2$  transport-matchable targets by interposing the map-transfer Lemma 5.1, the flow-approximation Lemma 5.4, and the coupling bound Lemma 5.2. Every node here is stated type-correctly in Lean against the axiom layer and left as a **sorry** (status open), except Proposition 6.2 — the generic  $O(1)$ -switch disentanglement heuristic — which we reviewed on paper only.

### 6.1 Lemma 5.1 — transport maps preserved by disentanglement

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.1, p. 24).

Suppose that for every  $i$  there exists  $T^i \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$  with  $T_{\#}^i \mu_0^i = \mu_1^i$ . Consider the flow map  $\Phi_1 : \mathcal{P}(\mathbb{S}^{d-1}) \rightarrow \mathcal{P}(\mathbb{S}^{d-1})$  (resp.  $\Phi_3$ ) given by Proposition 3.1 with data  $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$  (resp.  $\mu_1^i \in \mathcal{P}(\mathbb{S}^{d-1})$ ). Then there exists a Lipschitz-continuous and invertible map  $\psi : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  such that

$$\psi|_{\text{supp } \Phi_1(\mu_0^i)} = \psi^i|_{\text{supp } \Phi_1(\mu_0^i)} \quad (5.1)$$

for any  $i \in \{1, \dots, N\}$ , where  $\psi^i : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  is another Lipschitz-continuous and invertible map that satisfies

$$(\psi^i \circ T_{\Phi_1}^i)_\# \mu_0^i = (\psi \circ T_{\Phi_1}^i)_\# \mu_0^i = (T_{\Phi_3}^i)_\# \mu_1^i = \Phi_3(\mu_1^i)$$

for some Lipschitz-continuous and invertible maps  $T_{\Phi_1}^i, T_{\Phi_3}^i : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ . The proof can be found in Appendix B.4.

#### Our formalization. open

The paper’s strategy (Geshkovski, Rigollet, Ruiz-Balet, App. B.4, p. 37) exploits disentanglement: once Proposition 3.1 has pushed the supports  $\text{supp } \Phi_1(\mu_0^i)$  pairwise disjoint, the per-pair transport maps  $T^i$  are carried through the flow as Lipschitz-invertible maps  $T_{\Phi_1}^i, T_{\Phi_3}^i$ , and the individual matchings  $\psi^i$  can be glued into one global Lipschitz-continuous, invertible  $\psi$  that agrees with each  $\psi^i$  on the  $i$ -th (now disjoint) support and realizes  $(\psi \circ T_{\Phi_1}^i)_\# \mu_0^i = \Phi_3(\mu_1^i)$ . We give a type-correct Lean **statement** as a **sorry** (open), abstracting the Lipschitz/geodesic bookkeeping into the matchability hypothesis: from a per-index family of transport maps we ask for a single **Bijective**  $\psi$  pushing every  $\mu_0^i$  to  $\mu_1^i$ . We do not claim a proof.

```
-- **Lemma 5.1** (transport map after disentanglement). If each disentangled pair is
  ↪ matchable, a
single bijective map matches them all. Stub ('sorry'): 'math.open'. -/
theorem lemma_5_1 {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
  (hmatch : ∀ i, ∃ Ti : Eucl d → Eucl d, (μ₀ i).map Ti = μ₁ i) :
  ∃ ψ : Eucl d → Eucl d, Function.Bijective ψ ∧ ∀ i, (μ₀ i).map ψ = μ₁ i := by
  sorry
```

[Statements/MidLevel.lean](#)

## 6.2 Lemma 5.4 — $L^2$ approximation by a flow map

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.4, p. 24).

Suppose  $\varepsilon > 0$  and  $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ . For every  $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ , there exists a Lipschitz-continuous and invertible map  $\psi_\varepsilon : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  induced by the solution map of (B.1), namely  $\Phi_{\theta_\varepsilon}^T(\mu) = (\psi_\varepsilon)_\# \mu$  for some piecewise constant  $\theta_\varepsilon : [0, T] \rightarrow \Theta$  with finitely many switches, such that

$$\|\psi - \psi_\varepsilon\|_{L^2(\mu)} \leq \varepsilon.$$

The proof of Lemma 5.4 is involved, so we postpone it to Appendix B.5.

#### Our formalization. open

The paper (Geshkovski, Rigollet, Ruiz-Balet, App. B.5, p. 38) shows any  $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$  is approximated in  $L^2(\mu)$ , to any tolerance, by a flow map  $\psi_\varepsilon$  of the continuity equation (B.1) with piecewise-constant control and finitely many switches; chained with the coupling bound  $W_2(T_\#^1 \mu, T_\#^2 \mu) \leq \|T^1 - T^2\|_{L^2(\mu)}$  (Lemma 5.2, leaf L7) this converts  $L^2$  control of the map into  $W_2$  control of the pushforward. We give a type-correct Lean **statement** as a **sorry** (open):

for any  $\psi$  and  $\varepsilon$ , a parameter  $\theta$  and a map  $\psi_\varepsilon$  with  $\text{measureFlow } \theta \ T \ \mu = \mu.\text{map } \psi_\varepsilon$  and  $\sqrt{\int \|\psi - \psi_\varepsilon\|^2 d\mu} \leq \varepsilon$ . We do not claim a proof.

```

/-- **Lemma 5.4** ('L2' approximation by a flow map). Any transport map ' $\psi$ ' is
  ↪ approximated in
  ' $L^2(\mu)$ ' by a flow map of the dynamics, to any tolerance, with finitely many switches.
  ↪ Stub
('sorry'): ' $\text{math.open}$ '. Combined with the coupling bound (leaf L7) this controls ' $W_2$ '.
  ↪ -/
theorem lemma_5_4 (μ : Measure (Eucl d)) (ψ : Eucl d → Eucl d) (T ε : ℝ) (hT : 0 < T)
  ↪ (hε : 0 < ε) :
  ∃ (θ : Params d) (ψε : Eucl d → Eucl d),
    measureFlow θ T μ = μ.map ψε ∧ Real.sqrt (∫ x, ‖ψ x - ψε x‖ ^ 2 ∂μ) ≤ ε := by
    sorry

```

Statements/MidLevel.lean

### 6.3 Theorem 1.2 — matching transport-matchable targets

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.2, p. 5).

Suppose  $d \geq 3$ . Consider data  $(\mathcal{D})$  such that

- (1) there exist  $w_0, w_1 \in \mathbb{S}^{d-1}$  such that

$$w_0 \notin \bigcup_i \text{supp}(\mu_0^i) \quad \text{and} \quad w_1 \notin \bigcup_i \text{supp}(\mu_1^i); \quad (1.5)$$

- (2) for any  $i \in \{1, \dots, N\}$ , there exists  $T^i \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$  such that  $T_{\#}^i \mu_0^i = \mu_1^i$ .

Then for any  $T > 0$  and  $\varepsilon > 0$ , there exists  $\theta \in L^\infty((0, T); \Theta)$  such that for any  $i \in \{1, \dots, N\}$ , the unique solution  $\mu^i \in \mathcal{C}^0([0, T]; \mathcal{P}(\mathbb{S}^{d-1}))$  to (1.3) with data  $\mu_0^i$  and parameters  $\theta$  satisfies

$$W_2(\mu^i(T), \mu_1^i) \leq \varepsilon.$$

Moreover,  $\theta$  can be chosen piecewise constant.

*Proof sketch (full proof §5.1, p. 25).* The construction  $\Phi_{\text{fin}}^T := (\Phi_{\theta_3}^{T/3})^{-1} \circ \Phi_{\theta_2}^{T/3} \circ \Phi_{\theta_1}^{T/3}$  proceeds in three steps. **(1) Disentanglement:** Proposition 3.1 renders the input supports  $(\mu_0^i)$  pairwise disjoint via  $\Phi_{\theta_1}$  on  $[0, T/3]$  and the target supports  $(\mu_1^i)$  via  $\Phi_{\theta_3}$  on  $[2T/3, T]$ ; well-posedness and time-reversibility give a constant  $C$  with  $W_2((\Phi_{\theta_3}^T)^{-1}\mu, (\Phi_{\theta_3}^T)^{-1}\nu) \leq C W_2(\mu, \nu)$  (5.3). **(2) Matching:** Lemma 5.1 produces a single Lipschitz-invertible  $\psi$  agreeing with each  $\psi^i$  on the disentangled support, and Lemma 5.4 realizes  $\psi$  up to  $L^2$  error by a flow map  $\psi_\varepsilon = \Phi_{\theta_2}$  on  $[T/3, 2T/3]$ ; the coupling bound (Lemma 5.2) turns the  $L^2$  error into  $W_2((\Phi_{\theta_2}^{2T/3} \circ \Phi_{\theta_1}^{T/3})(\mu_0^i), \Phi_{\theta_3}^T(\mu_1^i)) \leq \varepsilon/C$  (5.5–5.6). **(3) Composition:** applying  $(\Phi_{\theta_3}^T)^{-1}$  and (5.3) yields  $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$  for all  $i$ .

### Our formalization. [open](#)

The paper’s proof (Geshkovski, Rigollet, Ruiz-Balet, §5.1) composes the three stages above: disentangle both families by Proposition 3.1, transfer the transport maps into a single  $\psi$  by Lemma 5.1, approximate  $\psi$  by a flow map by Lemma 5.4, and bound the resulting  $W_2$  error by the coupling Lemma 5.2. We give a type-correct Lean **statement** as a `sorry` (open). The hypothesis (1.5) — a shared missing direction off every input support and off every target support — and the  $L^2$ -matchability of the pairs are recorded as the opaque predicates `SharedMissingDirection` and `Matchable`, so the statement stays faithful without re-deriving the absent support / optimal-transport API; the conclusion is the controllability core, a single piecewise-constant  $\theta$  with  $W_2(\text{measureFlow } \theta \ T \ (\mu_0 \ i)) \ (\mu_1 \ i) \leq \varepsilon$  for every  $i$ . We do not claim a proof.

```

/-- **Theorem 1.2** (general targets). If every input/target pair is matchable by a
  ⇨ transport map
and the families share a missing direction, then for any horizon `T` and tolerance `ε`
  ⇨ there is a
piecewise-constant parameter `θ` whose solution map steers each input measure to within
  ⇨ `ε` of its
target in `W₂`. Stub (`sorry`): status `math.open`. -/
theorem theorem_1_2 (hd : 3 ≤ d) {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
  (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
  (hmiss₀ : SharedMissingDirection μ₀) (hmiss₁ : SharedMissingDirection μ₁)
  (hmatch : Matchable μ₀ μ₁) :
  ∃ θ : Params d, ∀ i, W2 (measureFlow θ T (μ₀ i)) (μ₁ i) ≤ ε := by
sorry

```

[Statements/MainResults.lean](#)

## 6.4 Theorem 1.1 — matching Dirac targets

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.1, p. 5).

Suppose  $d \geq 3$ . Consider data  $(\mathcal{D})$  such that

- (1) there exists  $w_0 \in \mathbb{S}^{d-1}$  such that

$$w_0 \notin \bigcup_i \text{supp}(\mu_0^i); \quad (1.4)$$

- (2) for any  $i \in \{1, \dots, N\}$ , we have  $\mu_1^i = \delta_{x^i}$ .

Then for any  $T > 0$  and  $\varepsilon > 0$ , there exists  $\theta \in L^\infty((0, T); \Theta)$  such that for any  $i \in \{1, \dots, N\}$ , the unique solution  $\mu^i \in \mathcal{C}^0([0, T]; \mathcal{P}(\mathbb{S}^{d-1}))$  to (1.3) with data  $\mu_0^i$  and parameters  $\theta$  satisfies

$$W_2(\mu^i(T), \mu_1^i) \leq \varepsilon.$$

Moreover,  $\theta$  can be chosen piecewise constant, with  $O(d \cdot N)$  switches, and

$$\|\theta\|_{L^\infty((0, T); \Theta)} = O\left(\frac{d \cdot N}{T} + \log \frac{1}{\varepsilon}\right).$$

*Proof sketch (full proof §5.2, p. 29).* The proof follows the same ideas as Theorem 1.2 but is significantly simpler — some steps are omitted — using  $\Phi_{\text{fin}}^T := \Phi_{\theta_3}^T \circ \Phi_{\theta_2}^{2T/3} \circ \Phi_{\theta_1}^{T/3}$ . **(1) Disentangle:**  $\Phi_{\theta_1}$  on  $[0, T/3]$  from Proposition 3.1, with  $\|\theta_1\| \lesssim dN/T$ . **(2) Cluster:**  $\Phi_{\theta_2}$  on  $[T/3, 2T/3]$  from Proposition 2.1, applicable since the  $\Phi^{T/3}(\mu_0^i)$  are now disjoint and single-hemisphere, driving each to within  $\delta$  of a point  $\delta_{x_0^i}$  with  $T_\delta = O(\log 1/\delta)$ , so  $\|\theta_2\| \lesssim \log 1/\delta$ . **(3) Match:**  $\Phi_{\theta_3}$  on  $[2T/3, T]$  from Proposition 4.1 steers the ensemble of atoms  $x_0^i$  to the targets  $y^i$ , with  $\|\theta_3\| \lesssim N/T$ ; since all measures are  $\delta$ -close to  $\delta_{x_0^i}$  the error is  $e^{O(N \cdot T)}\delta$ . All in all  $W_2(\mu^i(T), \delta_{y^i}) \leq \varepsilon$  with  $\|\theta\|_{L^\infty((0,T);\Theta)} = O(d \cdot N/T + \log 1/\varepsilon)$ .

### Our formalization. open

The paper’s proof (Geshkovski, Rigollet, Ruiz-Balet, §5.2) is the simpler Dirac-target specialization: disentangle by Proposition 3.1 ( $\|\theta_1\| \lesssim dN/T$ ), cluster each now-disjoint single-hemisphere measure to within  $\delta$  of a point by Proposition 2.1 ( $T_\delta = O(\log 1/\delta)$ ,  $\|\theta_2\| \lesssim \log 1/\delta$ ), then match the ensemble of cluster atoms  $x_0^i$  to the targets  $y^i$  by Proposition 4.1 ( $\|\theta_3\| \lesssim N/T$ ); Lemma 5.4 is not needed. The end-to-end construction is validated numerically by experiment **E6**, while the negative control **E7** shows that a single *linear* continuity equation cannot achieve it — motivating the measure-dependent vector field. We give a type-correct Lean **statement** as a `sorry` (open): a shared missing direction and Dirac targets  $\delta_{x^i}$  yield a single piecewise-constant  $\theta$  with  $W_2(\text{measureFlow } \theta \text{ } T (\mu_0 \text{ } i)) (\text{Measure.dirac } (x \text{ } i)) \leq \varepsilon$  for every  $i$ . The  $O(dN)$  switch count and the norm bound  $O(dN/T + \log 1/\varepsilon)$  are documented in this blueprint prose, not encoded in the Lean type. We do not claim a proof.

```
-- **Theorem 1.1** (Dirac targets). If the targets are point masses ' $\delta_{\{x \ i\}}$ ' and the
↪ inputs share
a missing direction, then for any horizon and tolerance a single piecewise-constant ' $\theta$ '
↪ steers each
input to within ' $\varepsilon$ ' of its target in ' $W_2$ '. Stub ('sorry'): status 'math.open'. The proof
↪ also gives
' $O(d \ N)$ ' switches and the stated norm bound (see the blueprint). -/
theorem theorem_1_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (x : Fin N →
↪ Eucl d)
  (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε) (hmiss : SharedMissingDirection μ₀) :
  ∃ θ : Params d, ∀ i, W2 (measureFlow θ T (μ₀ i)) (Measure.dirac (x i)) ≤ ε := by
sorry
```

[Statements/MainResults.lean](#)

## 6.5 Proposition 6.2 — generic $O(1)$ disentanglement

**From the original paper** (Geshkovski, Rigollet, Ruiz-Balet, Proposition 6.2, p. 30).  
Let  $d \geq 3$ ,  $\pi = \text{Unif}((\mathbb{S}^{d-1})^n)$ , and sample  $\mu_0^1, \dots, \mu_0^N$  i.i.d. from

$$\mathcal{S}_n := \left\{ \frac{1}{n} \sum_{k=1}^n \delta_{x_k} : x_k \in \mathbb{S}^{d-1} \right\} \simeq (\mathbb{S}^{d-1})^n.$$

Consider (1.3)–(1.2) with  $\mathbf{B} \equiv \beta I_d$  ( $\beta \geq 0$ ),  $\mathbf{V} \equiv I_d$ ,  $\mathbf{W} \equiv 0$ . Let  $x_*^i$  be the (a.s. existing) limit cluster point of  $\mu^i(t)$ . Then  $\mathbb{P}[x_*^i \neq x_*^j] = 1$  for  $i \neq j$ .

**Our formalization.** informal

**Reviewed on paper; not stated in Lean.** The paper’s argument (sketch of proof, p. 30) is a symmetry/measure-theoretic one: for  $\pi$ -almost every empirical input the dynamics with  $\mathbf{B} \equiv \beta I_d$ ,  $\mathbf{V} \equiv I_d$ ,  $\mathbf{W} \equiv 0$  converge to a Dirac  $\delta_X$ , defining a map  $f : \mathcal{S}_n \rightarrow \mathbb{S}^{d-1}$ . Because the flow and the law  $\pi$  are  $\mathcal{O}(d)$ -equivariant, the law  $\nu = f_{\#}\pi$  of the limit is rotation-invariant, hence non-atomic — an atom at  $x_0$  of mass  $p$  would force an atom of mass  $p$  at every  $Rx_0$ , contradicting  $\nu(\mathbb{S}^{d-1}) = 1$  once enough distinct rotations are chosen. Two independent inputs then give independent limits with this common non-atomic law, so  $\mathbb{P}[x_*^i = x_*^j] = \int \nu(\{x\}) d\nu(x) = 0$ . We do not state this in Lean: it would require Haar measure on  $\mathcal{O}(d)$ , the equivariant  $\omega$ -limit map  $f$ , and non-atomicity infrastructure outside our current axiom layer. The result is recorded as the heuristic that the generic switch complexity is  $O(1)$  rather than the  $O(dN)$  of Theorem 1.1.